

GPU-based SPH-DEM Method to Examine the Three-Phase Hydrodynamic Interactions between Multiphase Flow and Solid Particles

Young Beom Jo^a, So-Hyun Park^b, Hee Sang Yoo^a, Eung Soo Kim^{a,*}

^a Department of Nuclear Engineering, Seoul National University, 559 Gwanak-ro, Gwanak-gu, Seoul, South Korea

^b Korea Atomic Energy Research Institute, 989-111 Daeduk-daero, Daejeon, 305-353 Korea

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ABSTRACT

Particulate flows in a gas-liquid multiphase mixture (i.e., three-phase flows) are frequently encountered in various fields as well as the chemical engineering, resource engineering, ocean engineering, and nuclear engineering domains. Thus, the numerical modeling for such three-phase flows is essential to evaluate and enhance the fluidic design of the existing industrial devices or increase the engineering safety/efficiency. In terms of the solid particle phase, the discrete element method (DEM) has demonstrated outstanding capability in numerically modeling solid-solid interactions. For multiphase CFD model, several numerical techniques, ranging from the Eulerian-Eulerian two-fluid model to a fully resolved DNS solver, have been coupled with the DEM to model the three-phase flow. Most of the existing research on three-phase particulate flows was based on coupled grid-based CFD and DEM techniques. Considering that many three-phase particulate flows in industries are driven by a vapor/gas phase that forms a sharp interface with the liquid, coupling DEM with particle-based Lagrangian CFD methods, which can handle multi-phase flows without any interface capturing or associated correlations, can also be a good option or alternative for modeling such three-phase flows. In this respect, a three-phase flow solver was established in this study by coupling the DEM with a Lagrangian smoothed particle hydrodynamics (SPH) model while adopting the unresolved phase coupling between the two techniques. In order to overcome the low computational efficiency of these two numerical methods, GPU-based acceleration is adopted for speedup of SPH-DEM coupled solver due to its high compatibility for Lagrangian methods offered by thousands of GPU processors. The results of a scalability analysis indicated that the accelerated version exhibited an enhanced performance especially when many Lagrangian nodes are used in simulation. Several benchmark simulations were conducted to demonstrate the capability and applicability of the SPH-DEM solver in various multi-phase applications associated with particulate flows. The results of numerical simulations validated the feasibility of the momentum exchange model, and the potential and applicability of the SPH-DEM coupled solver for three-phase flow was demonstrated through a three-phase SPH-DEM simulation on the gas-driven leveling behavior of a particulate bed.

1. Introduction

The particulate flows in a gas-liquid mixture (i.e., three-phase flows) have attracted considerable research attention due to their widespread application. Such flows are ubiquitous in various fields such as those of chemical engineering, ocean engineering, resource engineering, nuclear engineering, and mechanical engineering. In particular, in the field of chemical and resource engineering, various complicated phenomena may occur due to the presence of gas bubbles in slurry reactors (An et al., 2020; Z. Peng et al., 2021; R. Zhou et al., 2017) and fluidization systems (Yu et al., 2020; Y. Zhou et al., 2020). Furthermore, various three-phase

flow issues are associated with the presence of fragmented debris particles in the field of nuclear reactor safety analysis. Specifically, two-phase natural convection flow often occurs through the packed debris due to the decay heat generated from debris particles (Hardee & Nilson, 1976; Sinha et al., 2008); this phenomenon leads to the leveling or sloshing behavior of the debris bed, in which the shape of the bed changes according to the two-phase flow rate (Cheng et al., 2013; B. Zhang et al., 2010). In such three-phase flows, complex physicochemical interactions occur among the multiple phases along with various interface behaviors.

The accurate numerical modeling for such three-phase flows is essential to evaluate and enhance the fluidic design of existing industrial

* Corresponding author.

E-mail address: kes7741@snu.ac.kr (E.S. Kim).

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Nomenclature		Subscript	
C_d	drag coefficient	a	targeted DEM particle
S_f	reaction force acting on SPH particle (SPH-DEM mtm exchange)	b	contacted or neighboring DEM particles
c_0	reference speed of sound [m/s]	c	contact point between DEM particles (DEM)
f_{eb}	external body force [m/s ²]	c	coupling domain (SPH-DEM)
$f_i(r)$	arbitrary field function	d	damper
f^n	magnitude of the normal contact force [N]	f	fluid
f^t	magnitude of the tangential contact force [N]	i	targeted SPH particle
$\hat{n}$	normal unit vector	j	neighboring SPH particles
$\hat{t}$	tangential unit vector	n	normal component
h	smoothing length [m]	s	spring
E	young's modulus [Pa]	t	tangential component
F	contact force between solid particles [N]	<i>Greek</i>	
G	shear modulus [Pa]	μ_r	resistive rolling friction coefficient
I	rotational inertia [kg·m ²]	μ_s	static friction coefficient
M	mass of DEM particle [kg]	μ_t	friction coefficient
R	radius of DEM particle [m]	ρ_{ref}	reference density [kg/m ³]
Re	Reynolds number	$\bar{\rho}$	superficial density [kg/m ³]
V	volume [m ³]	γ	polytrophic constant
W	kernel weighting function [m ⁻³]	δ	overlap between DEM particles [m]
d	dimension	ε	local porosity
e	restitution coefficient	κ	curvature [m ⁻²]
m	mass of SPH particle [kg]	λ	color function
p	pressure [Pa]	μ	dynamic viscosity [Pa·s]
v	velocity of DEM particles [m/s]	ν	kinematic viscosity [m ² /s]
g	gravitational acceleration [m/s ²]	ρ	density [kg/m ³]
n	normal vector [m ⁻¹]	σ	surface tension [N/m]
r	distance vector [m]	ω	angular velocity [rad/s]
u	velocity [m/s]	ϑ	pressure gradient [Pa/m]

devices or increase the engineering safety. Nevertheless, such modeling is challenging as the particle–particle collisions, fluid–particle interactions, and fluid–fluid interfacial behaviors must be considered at the same time. The traditional numerical research on three-phase flows was primarily realized using a macroscopic Eulerian–Eulerian continuum approach, in which the three phases are treated as a continuous phase (Buck, 2007; Grevskott et al., 1996; Yakush & Kudinov, 2011). Moreover, the conservation equations were solved in an averaged form based on a two-fluid model (TFM) with proper empirical constitutive equations. This approach could considerably reduce the computational burden; however, such approaches may overlook several major phenomena occurring in industrial devices, leading to improper modeling of dense solid beds (X. S. Sun & Sakai, 2015).

To overcome these numerical difficulties associated with solid particles, various approaches such as the discrete element method (DEM) (Cundall & Strack, 1979) and discrete particle model (DPM) (Deen et al., 2007) were established. In these approaches, the particulate solids are treated as a discrete phase, the behavior of individual particles is tracked in a Lagrangian framework, and the collisions among the solid particles are discretely solved based on appropriate contact force models. The numerical strategy of these methods enables the precise evaluation of the behavior of each solid particle. Notably, the outstanding capability of the DEM for simulating particulate flows has been demonstrated in many industrial fields.

In this respect, to model three-phase flows, the DEM is generally coupled with suitable gas–liquid multiphase CFD solvers. The approaches for coupling the CFD and DEM can be classified into two categories: resolved and unresolved coupling methods. In the former approach, the fluid and solid phases are completely resolved in the computational domain, and the fluid–solid interaction models involving

the surface tension force and surface wetting are directly solved. Thus, the resolved method allows for a more fundamental and practical coupling between the fluid and solid. This method is particularly useful when the phenomena is focused on the behavior of each individual solid particle or the flow disturbance by a small number of particles (Fujita et al., 2015; Mao et al., 2020). However, to model a particulate flow or packed bed, a large number of solid particles are required for the simulation, and in such scenarios, the computational load to fully resolve the fluid region between the solid particles may be extremely high. In such porous media, the unresolved coupling technique can be an effective alternative, in which the volume-averaged governing equations are locally solved for CFD models considering local porosity, and the inter-phase interactions are considered through semi-theoretical correlations or empirical force correlations. In the unresolved coupling approach, the capillary effect due to the flow path constriction and surface wettability is considered in the surface tension estimation, and solid particles experience the effect of the surface tension through the momentum exchange with local fluid. Although the specific flow information cannot be obtained, the unresolved coupling techniques are widely applied in practical systems due to its high capability for simulating debris flow (Li et al., 1999; Zhang et al., 2000; Jing et al., 2016; Li & Li, 2018). Each CFD models can be coupled with the DEM model both in resolved and unresolved manner depending on the characteristics of target phenomena.

The CFD methods for liquid–gas two-phase flows can be classified into three approaches: 1) TFM, 2) grid-based model, and 3) mesh-free Lagrangian model. In the TFM, the governing equations of the gas and liquid are solved in a locally averaged form involving suitable momentum exchange functions. By assuming the gas and liquid phase as continuous, the TFM model can decrease the computational burden.

However, it cannot provide the exact solution for the phenomena as gas–liquid interactions are based on empirical assumptions and correlations. Moreover, in terms of the CFD-DEM coupling, ensuring resolved coupling is essentially impossible in the TFM, and it is challenging to balance the momentum exchange among the three phases (X. S. Sun & Sakai, 2015; WEN et al., 2005).

To address the inherent difficulties of the TFM model, several direct numerical approaches have been developed to describe the motions of gas bubbles, such as the volume-of-fluid method (Hirt & Nichols, 1981), and level set method (D. Peng et al., 1999). These interface-capturing methods construct a geometrical representation of the interface or solve a transport equation for the indicator function to implicitly determine the location of the free surfaces or interfaces (Washino et al., 2021). Such interface-capturing methods are commonly used for numerically solving a gas–liquid–solid three-phase flow in a resolved (X. Sun & Sakai, 2016) or unresolved (Pozzetti & Peters, 2018) manner. However, in the case of resolved coupling with DEM, significant numerical effort may be required because the grid regeneration should be performed according to the movement of the solid particles.

In recent years, meshless Lagrangian methods, which can fundamentally address the interfacial interactions between bubbles and liquid, have attracted considerable attention, for instance, in the form of smoothed particle hydrodynamics (SPH) (Gingold & Monaghan, 1977; Lucy, 1977) and moving particle semi-implicit (MPS) (Koshizuka & Oka, 1996) techniques. Notably, the interfaces between the gas and liquid phases are traced without any transport equation due to the Lagrangian nature of particle-based meshless methods. In other words, the two different fluid phases are completely resolved, and the spatial distribution of the Lagrangian particles in each phase refers to the region of each phase. Thus, particle-based methods have been widely used in various gas–liquid multiphase flow applications (Chen et al., 2020; Liu et al., 2021; Ming et al., 2017). Moreover, the feasibility of Lagrangian CFD-DEM coupling is potentially high for both resolved and unresolved manner because two numerical models are both based on a Lagrangian methodology. Despite the high potential for phase coupling, however, the computational load was significantly high for both Lagrangian CFD and DEM methods before the dramatic improvement in the multicore hardware capability. Consequently, the coupling between particle-based CFD and DEM methods has been extensively considered only recently, and in particular, the numerical modeling of gas–liquid–solid three-phase flows have not been implemented for particle-based CFD methods.

In recent years, gas–liquid–solid three-phase flows have been modeled using a wide range of numerical techniques, from the Eulerian–Eulerian TFM to a fully resolved DNS solver. The CFD-DEM coupling studies so far have mostly focused on the coupling between grid-based CFD method and DEM taking account of computational efficiency, especially for gas–liquid–solid three-phase flows. However, considering that many three-phase particulate flows are driven by a vapor/gas phase that forms a sharp interface with the liquid, Lagrangian CFD methods, which can address multiphase flows without correlations or interface-capturing, can serve as an effective numerical approach to model gas-driven three-phase flows if they are properly accelerated. Considering these aspects, a parallelized three-phase flow solver was constructed in this study by coupling the DEM with a multiphase SPH model. The SPH-DEM coupling was performed in an unresolved manner to be suitable for numerous applications associated with the particulate flow and packed bed systems. Considering the numerical characteristics of SPH and DEM methods that follows the Lagrangian nature, the proposed SPH-DEM coupled solver was parallelized using a graphical process unit (GPU) with thousands of multiple cores. The performance of GPU parallelization was quantitatively evaluated through a simple scalability analysis for a benchmark simulation case.

This paper describes the numerical models and benchmark simulations of the proposed SPH-DEM coupled solver for multi-phase flow applications involving solid particles. The following two sections

present the governing equations for the SPH/DEM model and SPH-DEM coupled solver. The adopted GPU parallelization techniques are summarized in section 4, and the V&V results for benchmark problems performed to highlight the potential applications are discussed in section 5. Also, further required works for additional validation and practical application of proposed SPH-DEM model are introduced in section 6 with the brief summary of this study.

2. Numerical model description

2.1. Multiphase flow

2.1.1. SPH overview

Smoothed Particle Hydrodynamics (Gingold & Monaghan, 1977; Lucy, 1977; Monaghan, 1992) is a mesh-free Lagrangian method, which is particularly useful in modeling complex multi-phase, multi-physics flows in various engineering fields. The SPH is a numerical method that adopts a series of Lagrangian particles that carry individual properties to represent the fluid system, as shown in Fig. 1. The governing equations of each SPH particle are solved in a weighted averaged formulation over the particles in the support domain. The SPH approximation is based on an interpolation method using the Dirac-delta function $\delta(x)$:

$$f(x) = \int_{\Omega} f(x') \delta(x - x') dx' \quad (1)$$

where variables x and x' represent the points in the support domain Ω and integral domain, respectively, while $f(x')$ denotes a field function at x' . To determine the approximation of the function $f(x)$ in the SPH particle system, a kernel weighting function is introduced instead of the Dirac-delta function (Fig. 1), using which, Eq. (1) can be rewritten in terms of the kernel-weighted discretized formulation:

$$f_i(r) \approx \sum_j f_j W(r_i - r_j, h) V_j \quad (2)$$

where the subscripts i and j denote the target SPH particle and other SPH particles within the support domain of particle i , respectively, and $W(r_i - r_j, h)$ is the kernel weight function, with h representing the influencing area of the kernel function. The kernel weighting function is an approximation of the Dirac-delta function normalized over its support domain.

The SPH approximations of the function derivatives can be formu-

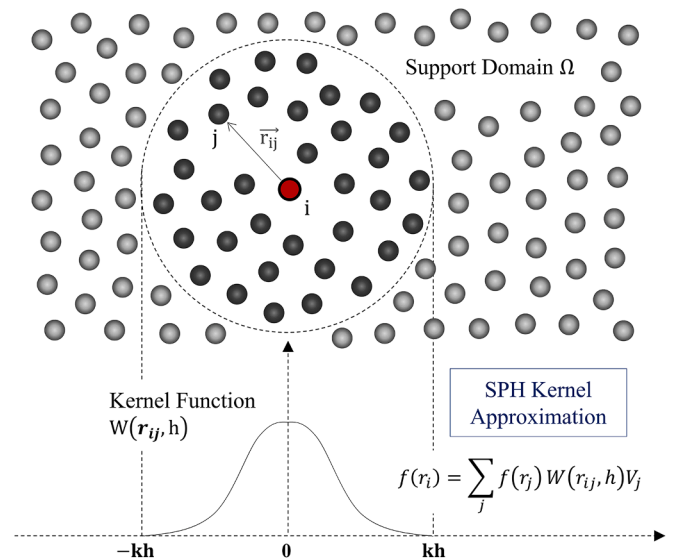


Fig. 1. Particle system and kernel weighting function in SPH method.

lated by applying the Gauss integral formula:

$$\nabla \cdot f_i(r) \approx \int_{\Omega} \nabla \cdot (f(r)W(r-r',h))dr' - \int_{\Omega} f(r)\nabla W(r-r',h)dr' \quad (3)$$

where the first term on the RHS of Eq. (3) can be replaced by the surface integral of the function $f(r)$ over the support domain (divergence theorem). Due to the compact condition of the kernel function, according to which the function has a distance-weighted value only in the interior area of the support domain (zero otherwise), the integral of the kernel function over the support domain surface becomes zero. Thus, the approximation of the first derivative can be simply summarized as a function of the kernel derivative:

$$\nabla \cdot f_i(r) = \sum_j f_j \nabla W(r_i - r_j, h) V_j \quad (4)$$

The SPH governing equations for the mass, heat, and fluid flow can be derived in several different forms by applying various differential quotient or product expansion to the basic SPH differential operator (Eq. (4)).

2.1.2. SPH discretized governing equations

The governing equations describing the fluid dynamics are the continuity equation and momentum equation, specifically, the Navier–Stokes equation. These two equations for incompressible Newtonian flow can be expressed in the Lagrangian forms as

$$\frac{d\rho}{dt} = -\rho \nabla \cdot u \quad (5)$$

$$\frac{du}{dt} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 u + g + f_{eb} \quad (6)$$

where ρ , p , and ν denote the density, pressure, and kinematic viscosity, respectively, and the vectors u , g , and f_{eb} represent the flow velocity, gravitational acceleration, and external body force, respectively. In the SPH method, these main governing equations are constructed in discretized form based on various kernel differential operators. The continuity equation can be discretized as follows:

$$\left(\frac{d\rho}{dt}\right)_i = \rho_i \sum_j \frac{m_j}{\rho_j} (u_i - u_j) \cdot \nabla_i W_{ij} \quad (7)$$

where u_i and u_j denote the velocity vector of the target particle and neighboring particle, respectively. In order to minimize the high frequency pressure noises generated by solving above continuity equation, several numerical techniques including delta SPH (Molteni & Colagrossi, 2009), deltaplus SPH (Sun et al., 2017), and Riemann solver (C. Zhang et al., 2017) are implemented in the SPH model of this study.

Meanwhile, the density of the targeted particle can also be estimated directly by substituting the normalized density into the field function of Eq. (2).

$$\left(\frac{\rho}{\rho_{ref}}\right)_i = \sum_j \frac{m_j}{\rho_j} \left(\frac{\rho_j}{\rho_{ref,j}}\right) W_{ij} / \sum_j \left(\frac{m_j}{\rho_j}\right) W_{ij} \quad (8)$$

where m and ρ_{ref} denote the mass and density in the reference thermodynamic state, respectively. Since the density of the targeted particle is estimated by the relative density to the reference density for each neighboring particles, the physical discontinuity of density is completely retained in a multi-phase interface regardless of the density ratio, as shown Fig. 2. The density discontinuity is well maintained in a two-fluid stratified interface without any averaging, and the gradient of hydrostatic pressure is also physically estimated without any smoothing. In this respect, the normalized density approach (equation (8)) is adopted in all simulations in this study due to its better capability for dealing multi-fluid & multi-phase flow than solving continuity equation

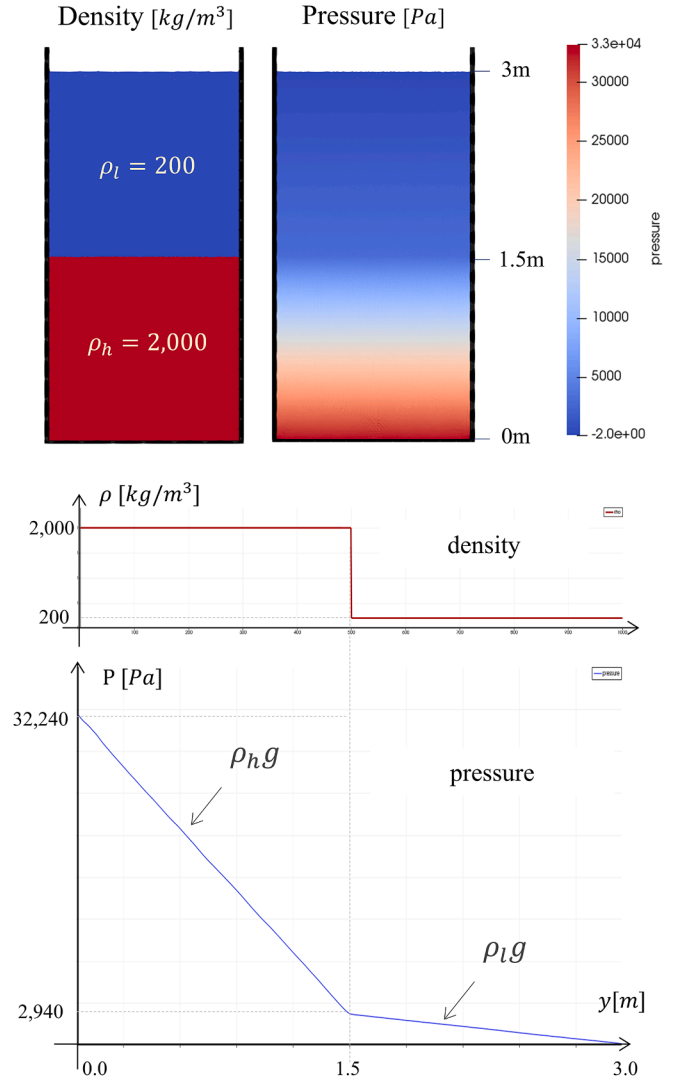


Fig. 2. Density and pressure field of two immiscible fluid interface.

(equation (7)).

Both density-estimation models Eqs. (7) and (8) assume the weak compressibility of the fluid particles and estimates the local pressure through the slight density change even for the incompressible fluid. In this respect, the standard SPH method is considered to be a weakly compressible SPH (WCSPH). The pressure field of each fluid particle is determined by the equation of state as a function of the density (Mongan, 1994):

$$p = \frac{c_0^2 \rho_{ref}}{\gamma} \left[\left(\frac{\rho}{\rho_{ref}} \right)^\gamma - 1 \right] \quad (9)$$

where γ denotes the polytropic constant that determines the sensitivity of the pressure calculation, and $c_0 = c(\rho_0)$ is the reference speed of sound. In the WCSPH approach, the density and momentum equations are closed by solving the state Eq. (9), rather than solving the partial differential Poisson equation for the pressure.

The momentum Eq. (6) in the discretized form is

$$\left(\frac{du}{dt}\right)_i = -\sum_j m_j \left(\frac{p_i + p_j}{\rho_i \rho_j}\right) \nabla W_{ij} + \sum_j \frac{m_j \mu_{ij} r_{ij} \cdot \nabla_i W_{ij}}{\rho_i \rho_j (|r_{ij}|^2 + \eta^2)} (u_i - u_j) + g - \frac{\sigma_i \kappa_i n_i + f_{eb}}{\rho_i} \quad (10)$$

where p , and μ denote the pressure and dynamic viscosity of the fluid, respectively. The pressure gradient force term in Eq. (10) (first term on the RHS) satisfies the linear momentum conservation inherently and ensures the physical validity of the estimated value even at multifluid/multiphase interfaces with a discontinuous density field. The laminar viscous force term in Eq. (10) (second term on the RHS) is derived by adopting the hybrid method expressing the second-order derivative (Monaghan, 1995). Since both terms in equation (10) are described as a function of the volume of neighboring particle j , it shows high capability in solving multi-phase flow (The volume of each particle is almost constant regardless of its density).

The surface tension force for small-scale multiphase flow can be expressed as a correlation equation for the curvature and interphase properties based on continuum surface force (CSF) model. The CSF model, which was originally derived through grid-based CFD methods, can estimate the surface tension effect on a macroscopic scale (Breinlinger et al., 2013; A. Zhang et al., 2015):

$$\left(\frac{du}{dt}\right)_i^{ST} = \frac{\sigma_i \kappa_i n_i}{\rho_i} \quad (11)$$

$$n_i = \sum_j \frac{1}{V_i} (V_i^2 + V_j^2) \lambda_{ij} \nabla W_{ij} \quad (12)$$

$$\kappa_i = -d \frac{\sum_j V_j (\hat{n}_i - \varphi_{ij} \hat{n}_j) \cdot \nabla W_{ij}}{\sum_j V_j |r_{ij}| |\nabla W_{ij}|} \quad (13)$$

$$\lambda_{ij} = \begin{cases} 0 & \text{for } f_i = f_j \\ \frac{\rho_i}{\rho_i + \rho_j} & \text{for } f_i \neq f_j \end{cases}, \quad \varphi_{ij} = \begin{cases} 1 & \text{for } f_i = f_j \\ -1 & \text{for } f_i \neq f_j \end{cases} \quad (14)$$

where σ , κ , n and d denote the surface tension coefficient, curvature, surface normal vector, and dimension ($d = 1, 2, 3$), respectively. λ_{ij} in Eq. (12) is a color function to identify different fluid phases, as defined in Eq. (14). In a three-phase flow region with small-scale solid particles, the curvature is estimated in a different form, which is described in section 3.2.

The benchmark simulations to realize the validation and verification of the constructed multiphase SPH model have been described in previous studies (Jo et al., 2019, 2021; Park et al., 2020).

2.2. Solid particle phase

2.2.1. Discrete element method (DEM)

Discrete Element Method is a particle-scale direct numerical method to model the bulk behavior of granular materials including translation, rotation, and interparticle collision (Cundall & St, 1979). The constructed DEM model regards each particle as a single-node element that

has a rigid spherical shape, and the collision between the spherical particles is solved considering a spring-dashpot system, which allows a thin overlap of the contact particle pair (Fig. 3). The spring-dashpot system consists of a spring, damper, and slider, as shown in Fig. 3, which provide the functionalities of elasticity, damping, and friction, respectively.

The model for the contact force between the DEM particles can be classified based on the description of the elastic force, for instance, in the form of a linear model (Misra & Cheung, 1999), nonlinear restoring model, Johnson-Kendall-Roberts (JKR) model, or other relevant models (Johnson et al., 1971). In this study, the nonlinear Hertz-Mindlin contact force model is adopted due to its accuracy and simplicity (B. Peng et al., 2014; Raji et al., 1999; Zhu et al., 2007). The collision force between the particles is divided into the normal and tangential components, which are further divided into the elastic and damping terms:

$$F_{ab} = f_{ab}^n \hat{n} + f_{ab}^t \hat{t} = (f_s^n + f_d^n) \hat{n} + (f_s^t + f_d^t) \hat{t} \quad (15)$$

where the unit vector $\hat{n}$ and $\hat{t}$ respectively denote the normal and tangential components between particles a and b. f_{ab}^n and f_{ab}^t represent the magnitude of the normal and tangential components of the contact force, respectively, and subscripts s and d indicate the spring and damper, respectively. In the Hertz-Mindlin contact model, each term on the RHS of Eq. (16) can be defined as follows:

$$f_s^n = \frac{4}{3} E_{ab}^* \sqrt{R_{ab}^*} \delta_n^{3/2} \quad (16)$$

$$f_d^n = -\sqrt{\frac{10}{3}} \frac{\ln(e)}{\sqrt{\ln(e)^2 + \pi^2}} \sqrt{M_{ab}^* K'_n} v_n \quad (17)$$

$$f_s^t = 8 G_{ab}^* \sqrt{R_{ab}^*} \delta_n \delta_t \quad (18)$$

$$f_d^t = \sqrt{\frac{10}{3}} \frac{\ln(e)}{\sqrt{\ln(e)^2 + \pi^2}} \sqrt{M_{ab}^* k_t} \quad (19)$$

where e is the restitution coefficient between particles a and b , δ_n and δ_t denote the interparticle overlap in the normal and tangential directions, respectively, and v_n and v_t indicate the relative velocity components in the corresponding directions. M_{ab} , R_{ab} , E_{ab} , and G_{ab} in Eqs. (16)–(19) represent the equivalent properties for the mass, radius, Young's modulus, and shear modulus, respectively.

The overlap in the normal direction (δ_n) can be simply calculated from the relation between the particle distance and radius of each particle, whereas the tangential overlap δ_t is not a geometrically determined variable. Instead, the tangential overlap can be determined by accumulating the relative velocity between the particles at the contact point in the collision period:

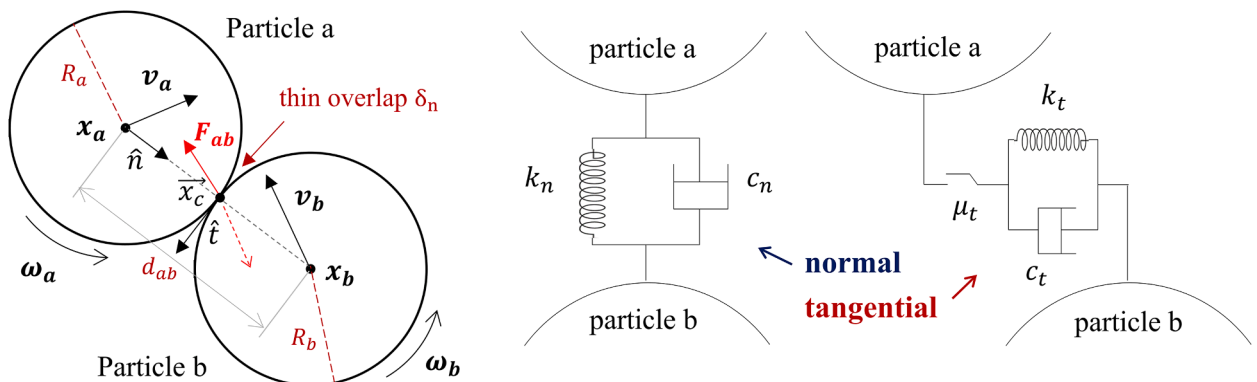


Fig. 3. Spring-dashpot system of soft-sphere DEM collision model.

$$\Delta \delta_t = v_t \Delta t' \quad (20)$$

where $\Delta t'$ is the time-step size of the DEM model. When the contact force exceeds the maximum friction force, sliding motion occurs between two contact particles. Thus, the contact force in the tangential direction can be written as follows:

$$f_{ab}' = \min(f_s', f_d', -\mu_t |f_{ab}' \hat{n}|) \quad (21)$$

where μ_t is the friction coefficient between the two contact particles.

When the contact forces in interparticle collisions are estimated through the abovementioned contact force models, the translational and rotational motion of each DEM particle can be determined according to Newton's law of motion:

$$m_a \frac{v_a}{dt} = \sum_b F_{ab} + m_a g \quad (22)$$

$$I_a \frac{\omega_a}{dt} = \sum_b (r_{ac} \times F_{ab}') \quad (23)$$

where m_a and I_a denote the mass and rotational inertia of particle a , respectively, and r_{ac} is the position vector from the center of particle a toward the contact point c .

2.2.2. Versatile wall boundary conditions

The interactions between the solid particles and wall boundary can be classified into two types: collision with a sufficient normal velocity component and surface motion (rolling or sliding). These two types of boundary motions are fundamentally based on different physical phenomena. For instance, solving the solid particle behavior based only on a collision dynamic may cause an abnormally large friction force, leading to the underestimation of the particle surface motion. To address this problem, a versatile wall boundary model to suitably select from between collision or surface motion is established, as shown in Fig. 4.

First, the contact detection of DEM particles is realized based on the geometrical function for the wall boundary, and the surface normal is calculated from the wall boundary information. Next, the collision or surface behavior is selectively determined based on the magnitude of the velocity component in the normal direction to the boundary surface, as shown in Fig. 4. Specifically, the governing equations for the collision dynamics are solved if the magnitude of the normal velocity component is sufficiently large:

$$v_1 = v_0 - (1 + e_n) (v_{c0} \cdot \hat{n}) \hat{n} - \frac{1}{m} \left(\frac{1}{m} + \frac{r^2}{I} \right)^{-1} (1 + e_t) (v_{c0} \cdot \hat{t}) \hat{t} \quad (24)$$

$$\omega_1 = \omega_0 + \frac{r}{I} \left(\frac{1}{m} + \frac{r^2}{I} \right)^{-1} (1 + e_t) (v_{c0} \cdot \hat{t}) (\hat{n} \times \hat{t}) \quad (25)$$

where v_0 and ω_0 denote the initial velocity and angular velocity vector, respectively, and v_{c0} represents the velocity vector at the contact point C, considering the rotational motion of the DEM particle. m and I denote the mass and rotational inertia, respectively, an e_n and e_t are the restitution coefficients of collision in each direction.

If the magnitude of the normal velocity component is below a certain threshold (i.e., continuous contact with the surface), a sliding or rolling equation is solved for each DEM particle:

$$v = v_0 + (\mu_s \hat{n} \cdot \hat{t}) \hat{t} \quad (26)$$

$$\omega = \omega_0 + \left(\frac{m(\mu_s + \mu_r) g \cdot \hat{n}}{I} r \times \hat{t} \right) \hat{t} \quad (27)$$

where μ_s and μ_r are the coefficients for static friction and resistive rolling friction, respectively, and g denotes the gravitational acceleration vector.

3. Fluid–solid three-phase interaction

3.1. SPH-DEM Unresolved Coupling

The unresolved coupling is realized between the multiphase SPH and DEM methods, which is more effective for engineering/industrial applications involving particulate flow or packed beds. In the unresolved coupling approach, the SPH particle spacing is comparable to the DEM particle diameter, and both particles exist in the same computational domain with a spatial overlap, as shown in Fig. 5. In general, the coupling domain must be set to be considerably larger than the DEM particle diameter to ensure an accurate momentum exchange. The coupling radius in all the simulations in this study is set to be at least six times the diameter of solid spheres to ensure that the behavior of each DEM particle can be estimated based on sufficient information of the local flow field.

3.2. Coupling equations

For the liquid and gas phases (SPH), the locally averaged Navier–Stokes equation is solved considering the spatial interaction with a solid phase (DEM) represented by the local porosity of fluid. The corresponding momentum equation can be written in the Lagrangian form as follows:

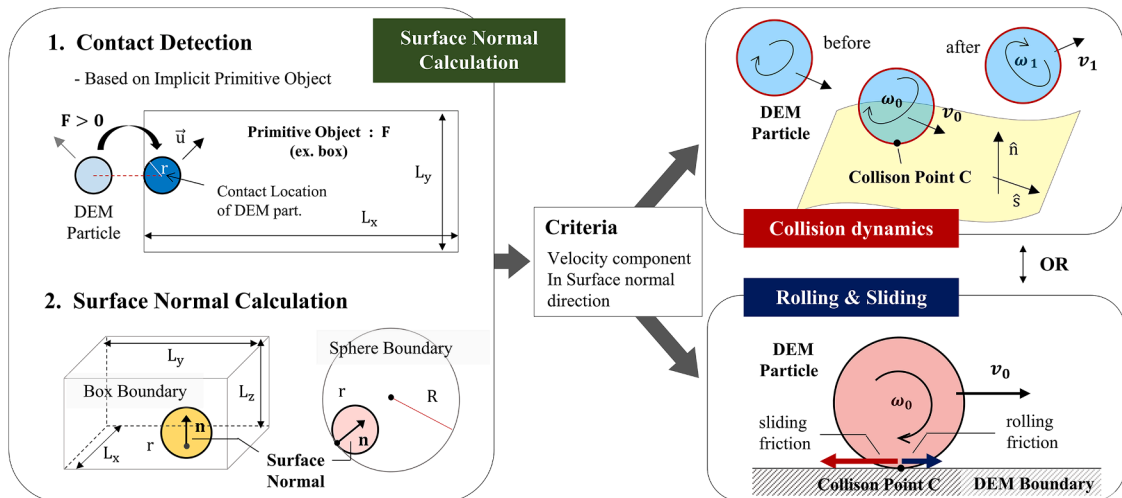


Fig. 4. Versatile wall boundary model.

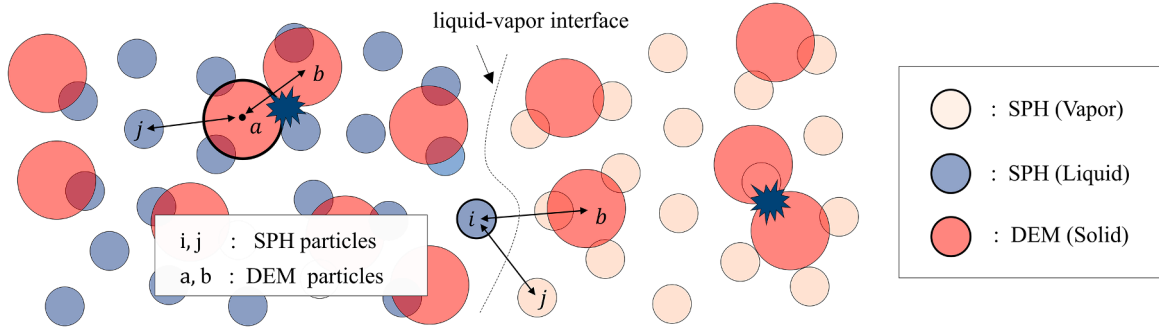


Fig. 5. Unresolved coupling domain of SPH and DEM.

$$\rho_f \frac{\partial(\epsilon u_f)}{\partial t} = -\epsilon \nabla P + \mu_f \nabla \cdot (\epsilon \nabla u_f) + \epsilon \rho_f g + \epsilon \sigma_i \kappa_i n_i + S_f \quad (28)$$

where ϵ is the local volume fraction of fluid f , and ρ_f and μ_f are the density and dynamic viscosity of the fluid, respectively. The momentum exchange term S_f is the reaction force acting on the fluid due to the hydrodynamic interaction with the local DEM particles within the coupling domain. The local porosity of the SPH particle i is calculated through a summation over the surrounding DEM particles within a coupling length h_c :

$$\epsilon_i = 1 - \left(\sum_b W_{ib}(h_c) V_b / \sum_j W_{ij}(h_c) V_j \right) \quad (29)$$

where subscripts i and j denote the target SPH particle and neighboring SPH particles, respectively, and subscript b refers to the neighboring DEM particles. The governing equation for the motion of SPH particles is organized in the form of a locally averaged Navier–Stokes equation:

$$\left(\frac{du}{dt} \right)_i = -\sum_j m_j \left[\frac{P_i}{\rho_i^2} + \frac{P_j}{\rho_j^2} \right] \nabla_i W_{ij} + \frac{\sigma_i}{\rho_i} \kappa_i n_i + \bar{\rho}_i g + S_i \quad (30)$$

where $\bar{\rho}$ denotes the superficial fluid density, which is defined as $\bar{\rho} = \epsilon \rho$. In the case of surface tension force, the macroscopic curvature is calculated based on Eq. (13) in a fluid region with few or no solid particles where $\epsilon_i \geq 0.8$. In the case of packed region ($\epsilon < 0.8$), the curvature is re-estimated as a function of contact angle θ in order to take into account the capillary action (Washino et al., 2013),

$$\kappa_i = -\frac{2 \cos \theta}{r_\epsilon} \text{ where } r_\epsilon = \frac{2\epsilon D_p}{3(1-\epsilon)} \quad (31)$$

where r_ϵ is the effective capillary radius.

The forces acting on each DEM solid particle are estimated from the local flow information within the coupling domain. In this study, the pressure gradient force and drag force are considered (Y. He et al., 2018):

$$F_a = V_i \sum_j \frac{m_j}{\rho_j} \vartheta_j W_{aj} / \sum_j \frac{m_j}{\rho_j} W_{aj} + \frac{1}{8} C_d \rho_f \pi d^2 |u_{ja}| u_{ja} f(\epsilon_a) \quad (32)$$

The first and second terms in the RHS of Eq. (32) correspond to the pressure gradient force and drag force, respectively. The pressure gradient force of the solid (DEM) particle is calculated by interpolating the pressure gradient (ϑ) of the surrounding flow field through the weight averaging of the neighboring SPH particle information. Considering the pressure jump occur at the three-phase flow interface dominated by surface tension, the kernel interaction for pressure gradient is performed only with the SPH particles to which each targeted solid particle belong, as shown in Fig. 6. In addition, since the first particle layer at the interface has numerically smoothed pressure field, the particles of the corresponding layer are also excluded from kernel interaction. Therefore, the kernel gradient correction is applied for θ estimation to consider only the particles included in the kernel interaction as follow,

$$\vartheta_i = -\bar{\rho}_i \sum_j \bar{m}_j \left[\frac{P_i + P_j}{\bar{\rho}_i \bar{\rho}_j} \right] L_i^{-1} \nabla_i W_{ij} \quad (33)$$

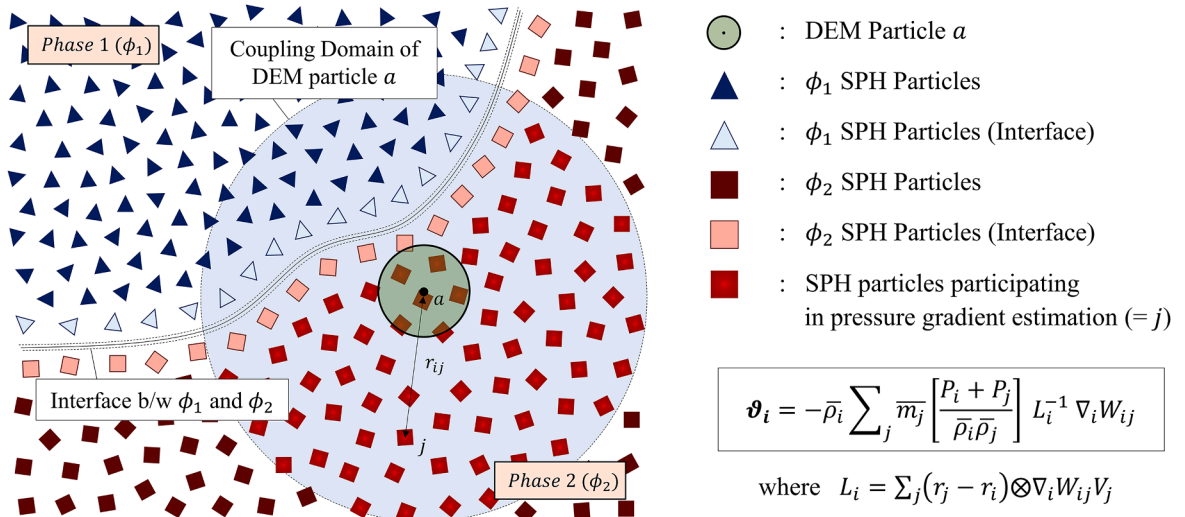


Fig. 6. Pressure gradient force estimation of DEM solid particle at the interface.

$$L_i = \sum_j (r_j - r_i) \otimes \nabla_i W_{ij} V_j \quad (34)$$

where the subscript j belongs to the neighbor SPH particles participating in the pressure gradient estimation as shown in Fig. 6. Solid particles at the liquid-vapor interface experience the effect of surface tension through this pressure gradient force. Specifically, when a solid particle is instantaneously subjected to a force to exit the phase boundary, the curvature of the interface, a concave curvature can be formed at the interface due to the drag caused by the jetted solid particle. In this case, this solid particle can be trapped by the drag of local restoring flow due to the surface tension force and the inward pressure gradient force acting on the solid particle as well.

The drag force of each DEM particle is estimated using a drag coefficient correlation, in which the term $f(\varepsilon_a)$, which is the correction factor reported by Di Felice (di Felice, 1994), is used to account for the behavior of the particulate system:

$$f(\varepsilon_a) = \varepsilon_a^{1-\xi} \text{ where } \xi = 2.65(\varepsilon_a + 1) - (5.3 - 3.5\varepsilon_a)\varepsilon_a^2 \exp\left(-\frac{(1.5 - \log(Re))^2}{2}\right) \quad (35)$$

where ε_a is the weighted volume-averaged fluid fraction around the target DEM particle a . The local Reynolds number in Eq. (35) is calculated as a function of a superficial relative velocity identified considering the local porosity. The drag force equation including the correction factor correlation (Eq. (35)) is validated for various Reynolds numbers and packing ratios, which is described in section 5.2.

Furthermore, the momentum exchange term S_f in the SPH governing equation (Eq. (28)) is calculated using a weighted average of the fluid-particle coupling force acting on the neighboring DEM particles:

$$S_i = -\frac{1}{\rho_i} \sum_b \frac{1}{\sum_j \frac{m_j}{\rho_j} W_{bj}} F_b W_{ib} \quad (36)$$

where the subscript j refers to the neighboring SPH particles for DEM particle b , and subscript b refers to surrounding DEM particles within the coupling domain of the targeted SPH particle i (Fig. 7). Thus, the momentum exchange between the fluid and solid phases satisfies Newton's third law, and the importance of this momentum exchange term is highlighted in the fluid-solid dam-break simulation described in section 5.3.

3.3. Coupling algorithm

Considering the time-step size required in two-phase SPH and DEM method, the physical models for SPH and DEM are implemented in a single algorithm as shown in Fig. 8.

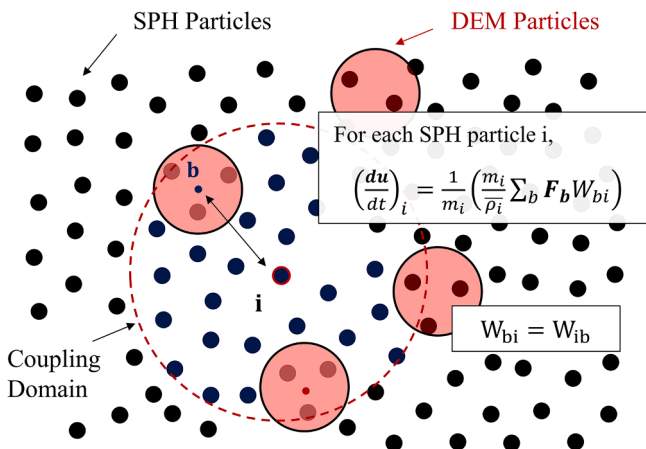


Fig. 7. SPH-DEM Momentum exchange in coupling domain.

The physical models for the SPH and DEM methods are identically solved at each Lagrangian point. Nearest neighbor particle search (SPH) and contact detection (DEM) are performed based on each grid cell, which is described in section 4.2 in detail. The mutual search between the SPH and DEM particles in the fluid-solid interaction step is realized based on the SPH cell configuration, as shown in Fig. 8.

A modified predictor-corrector scheme is applied to perform the time integration of the coupled SPH-DEM model. This method divides the time integration into two steps (i.e., predictor and corrector) and determines the physical variables (density, pressure, force, etc.) based on the predicted values. Specifically, the prediction step extrapolates the physical variables as a function of the initial values (t) and time derivatives in the previous time-step ($t - \Delta t/2$):

$$\begin{aligned} u_{t+\frac{\Delta t}{2}}^p &= u_t + \frac{\Delta t}{2} \left(\frac{du}{dt} \right)_{t-\frac{\Delta t}{2}} \\ r_{t+\frac{\Delta t}{2}}^p &= r_t + \frac{\Delta t}{2} \left(\frac{dr}{dt} \right)_{t-\frac{\Delta t}{2}} \\ h_{t+\frac{\Delta t}{2}}^p &= h_t + \frac{\Delta t}{2} \left(\frac{dh}{dt} \right)_{t-\frac{\Delta t}{2}} \end{aligned} \quad (37)$$

where t and Δt denote the initial time of the current step and time-step size, respectively, and the superscript p denotes the predictor step. The time derivatives of the position, velocity, enthalpy, and density (SPH) are estimated based on the predicted values by solving the governing equations for the SPH, DEM, and SPH-DEM interactions. Finally, the field variables are re-integrated using the newly estimated time derivatives:

$$\begin{aligned} u_{t+\Delta t}^c &= u_t + \Delta t \left(\frac{du}{dt} \right)_{t+\frac{\Delta t}{2}} \\ r_{t+\Delta t}^c &= r_t + \Delta t \left(\frac{dr}{dt} \right)_{t+\frac{\Delta t}{2}} \\ h_{t+\Delta t}^c &= h_t + \Delta t \left(\frac{dh}{dt} \right)_{t+\frac{\Delta t}{2}} \end{aligned} \quad (38)$$

where the superscript c denotes the corrector step.

4. GPU implementation

SPH and DEM is considered to be a computationally expensive method. The time-step size required for simulation is quite small ($< 1.0e^{-4}$) due to the numerical characteristics of the Weakly Compressible SPH method and DEM that follow the explicit time integration scheme. The smaller time-step size requires more iterations, which leads to an increase in the total computation time. Therefore, if the inter-particle interactions for all Lagrangian nodes (both SPH and DEM) are serially calculated, the application of this methodology can be crucially limited both in respect of length-scale and time scale. In addition, the accuracy of SPH method is greatly affected by particle resolution, especially in the case of simulations involving free surfaces or multi-fluid (phase) interfaces.

In these reasons, enhancing computational performance is essential to make this SPH-DEM method more practical and applicable. Recently, various High-Performance Computing (HPC) techniques have been actively employed to address the efficiency issue: OpenMP, MPI, GPGPU, etc. Since the parallel programming approaches are different for each HPC technique, an appropriate option must be adopted according to the numerical scheme used. In this study, GPU-based acceleration is adopted for speedup of SPH-DEM coupled solver due to its high memory bandwidth and instruction throughput offered by GPU processors, which are optimized for handling large numbers of Lagrangian particles

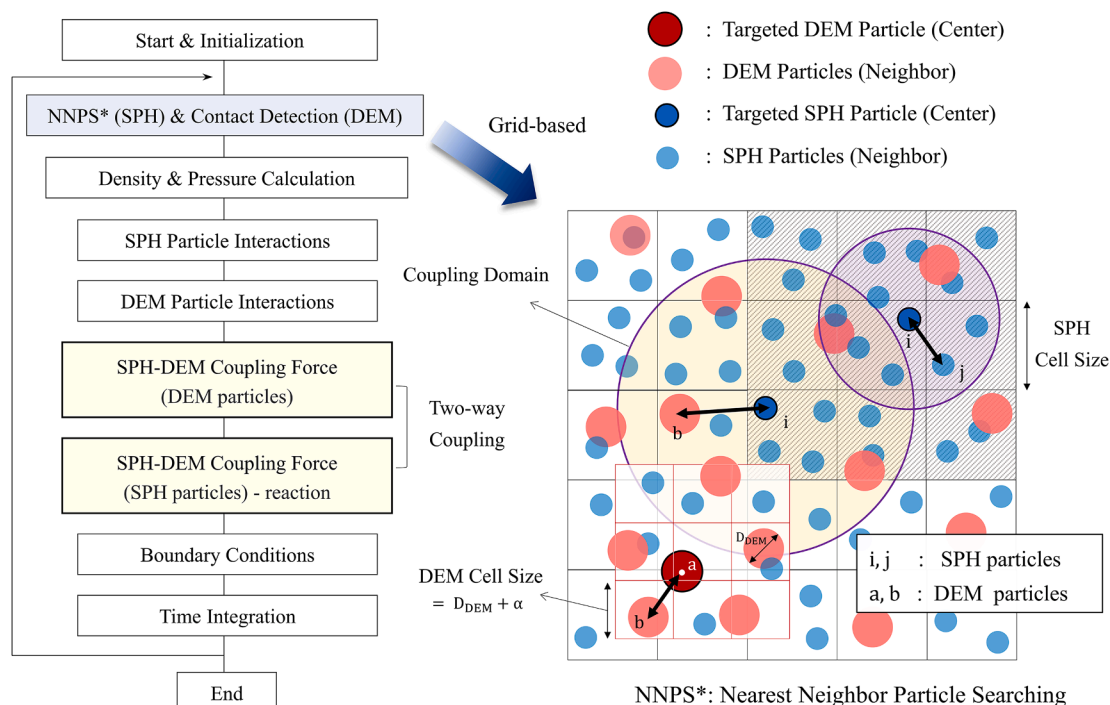


Fig. 8. Algorithm and cell configuration of SPH-DEM solver.

(SPH/DEM).

4.1. Memory management and algorithm

Fig. 9 shows the simplified structure of the GPU memory (Kirk, 2007). The GPU consists of multiple streaming multiprocessors (SM) composed of dozens of CUDA cores. The GPU parallelization of the SPH-DEM coupled model is implemented by allocating each single particle (SPH/DEM) to the corresponding GPU thread, which is the

smallest unit to conduct executions. These threads are executed in particular CUDA cores, whereas each SM concurrently handles the allocated blocks, each of which is a set of several threads.

Fig. 10 shows the simplified algorithm of the numerical model with the GPU acceleration. The initialization step (reading input, setting parameters, etc.) is implemented by CPU (host), whereas the other steps including the governing equation solvers are parallelized for each SPH/DEM particle using the GPU (device). The SPH and DEM particle data read in the initialization step are copied to GPU memory after the

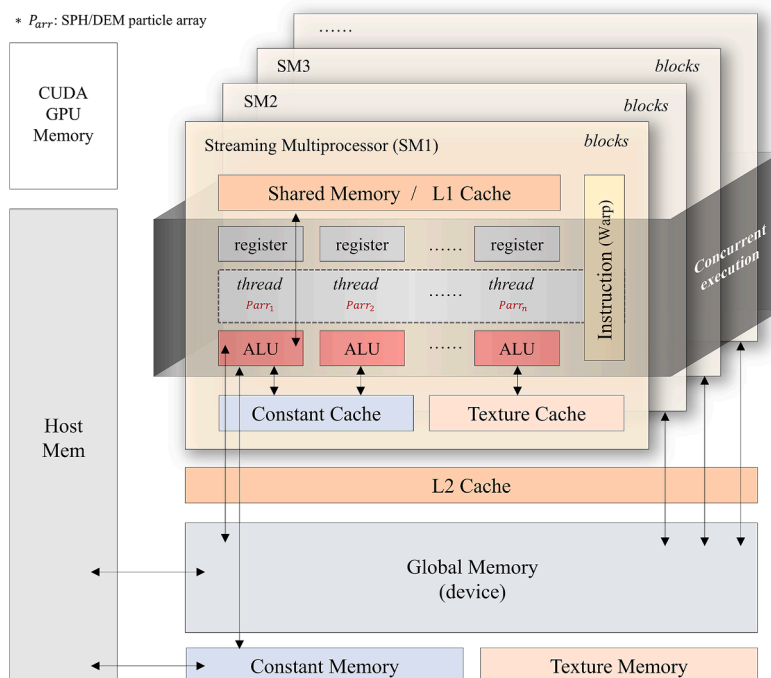


Fig. 9. CUDA GPU memory structure.

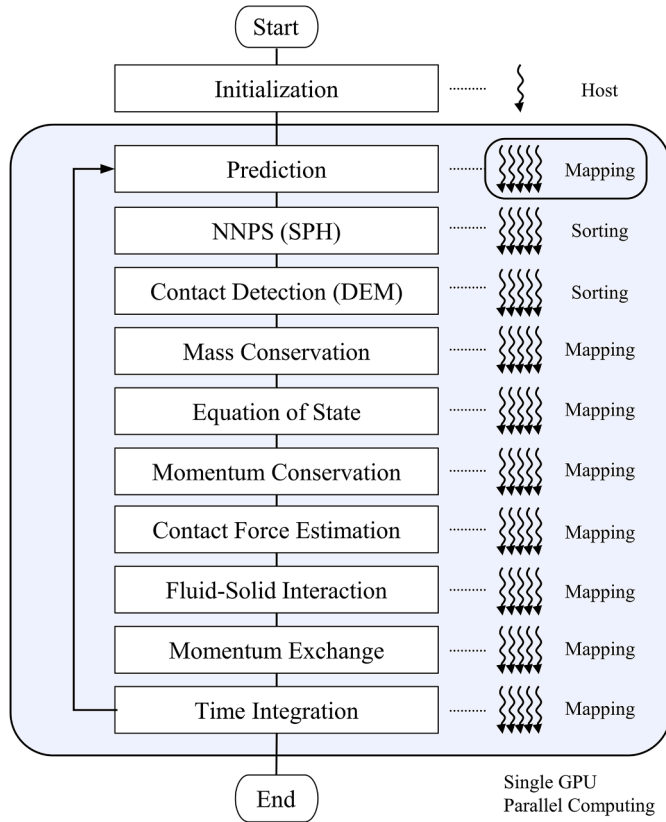


Fig. 10. Concurrent execution of parallelized SPH-DEM solver.

initialization step (host → device). Subsequently, the governing equations of each SPH/DEM particle are solved in parallel (GPU), and finally, the results of numerical simulation are copied to the CPU memory to generate the output data. The details of the concurrent execution for the SPH/DEM solver are provided in the following subsection.

4.2. Concurrent execution

The parallelization strategy of the SPH/DEM solver can be classified into two types. The primary strategy is to solve the governing equations

for each SPH/DEM particle concurrently, and the other strategy is to parallelize the sorting algorithm to improve the memory access efficiency of the SPH NNPS and contact detection between DEM particles.

The governing equations are solved in parallel for each SPH/DEM particle (parallel mapping), as shown in Fig. 10. Specifically, as shown in Fig. 9, each Lagrangian particle is allocated to a single GPU thread to ensure that all or part of the calculation is executed in parallel according to the GPU specifications and simulation conditions (number of particles, number of variables, etc.).

The SPH NNPS and DEM contact detection are implemented in parallel for each particle in the concurrent framework. However, an efficient additional strategy is required for these steps with respect to the memory access to neighbor particles or contact particles. The NNPS and contact detection are optimized using an algorithm based on uniform grid and parallel sorting, which rearranges each SPH/DEM particle according to the grid index order. The specific order of parallel sorting is as follows (Fig. 11):

- 1) The grid cell index (GI) of each particle is calculated.
- 2) The particle array (PI) is rearranged (PI^*) in the order of the grid cells (GI^*).

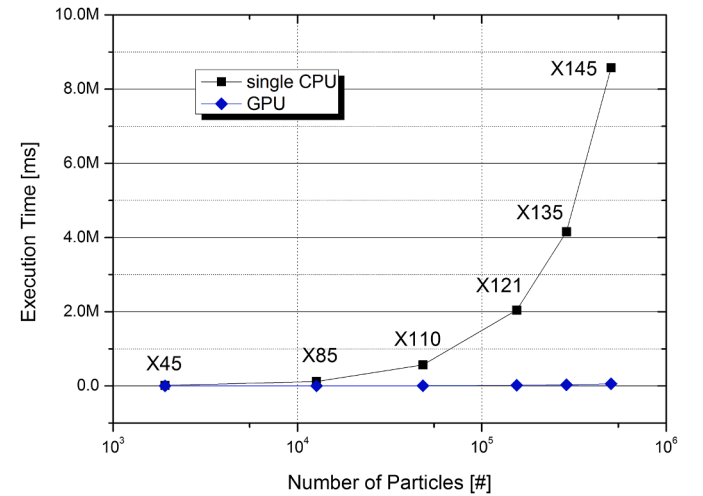


Fig. 12. Speedup factor for single GPU Parallelization.

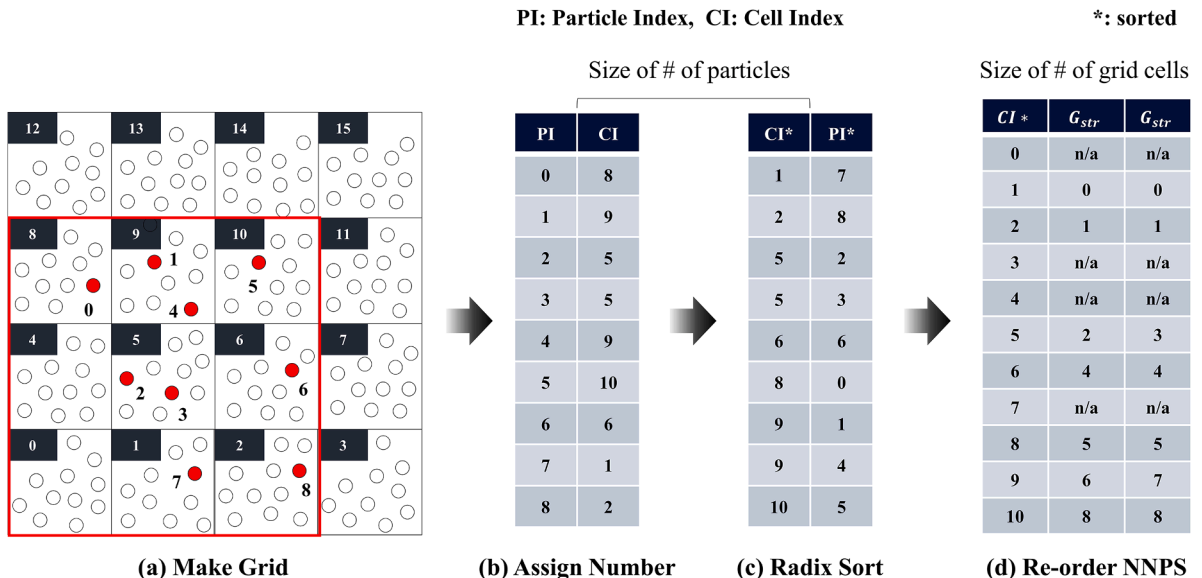


Fig. 11. Algorithm of NNPS and contact detection.

- 3) The cell arrays for the starting (G_{str}) and ending (G_{end}) particle indexes are constructed.

where * indicates the sorted array. Based on this sorted particle array, the NNPS and contact detection are performed for each targeted particle from the particle corresponding to the starting index (G_{str}) of each neighboring grid cell to that corresponding to the ending index (G_{end}). The SPH/DEM particles in the same grid cell exist in GPU memory close to one another, which helps increase the cache hit rate during the NNPS and DEM contact detection.

4.3. Performance evaluation

The performance of GPU parallelization can be quantified through scalability analysis. Since the DEM model and the SPH-DEM coupled model were numerically implemented in the already parallelized environment, the efficiency of GPU was evaluated by calculating the reference factors (speedup factor, parallelization factor, etc.) for 2D SPH dam-break simulation case (Jo et al., 2019). These reference factors were calculated for four different number of Lagrangian nodes as shown in Figure a. In summary, the speedup factor was calculated as 157.9 when using one million SPH particles, which means it has been more than 160 times faster than single-CPU calculation. In this case, the parallelization factor exceeded 0.99, which was expected to lead to a sufficient performance improvement. Furthermore, the speedup factor reached 96.3% of the convergence value, indicating the efficiency of the parallelization for the given conditions. Although these results corresponded to SPH modeling, a similar level of performance improvement can be expected for the SPH-DEM solver when considering the similarity of the algorithms of the SPH and DEM techniques.

5. Benchmark simulations and discussions

Various benchmark simulations are performed to validate the constructed SPH-DEM solver and evaluate its feasibility in modeling multiphase phenomena. This section presents several validation results and discussions for each relevant application.

5.1. Terminal velocity

The simplest coupling between SPH and DEM would be to analyze a single solid particle behavior within a single-phase flow. As the proposed SPH-DEM coupled model in this study targets the overall multiphase flow analysis accompanied by solid particles, it is necessary to evaluate a single particle behavior as a step for overall performance demonstration. Also, during the SPH-DEM coupled simulation, there is a possibility that isolated single particles may occur during the simulation, which are likely to be attributed to one specific phase. In these reasons, the terminal velocity behavior of a single solid particle is estimated with a brief sensitivity test. The numerically estimated velocity of a single DEM particle is tracked over time and compared with the analytical velocity determined using the following equation:

$$m_s \ddot{z} = -m_s g + m_s \frac{\rho_f}{\rho_s} g - \frac{\rho_f |\dot{z}|}{2} C_d \left(\frac{\pi d_s^2}{4} \right) \dot{z} \quad (39)$$

where the subscripts f and s refer to the fluid and solid, respectively. d_s and z denote the diameter of the solid particles and position vector in the direction opposite to that of gravity, respectively, and C_d indicates the drag coefficient. The time variation of the estimated velocity is in good agreement with the results obtained from the theoretically derived Eq. (39), as shown in Fig. 13. In this simulation, one-way phase coupling is performed, in which only a DEM particle receives the forces from the neighboring SPH particles, so that the drag force for the solid particle is solved for undisturbed fluid velocity.

The drag force model is estimated based on the difference between

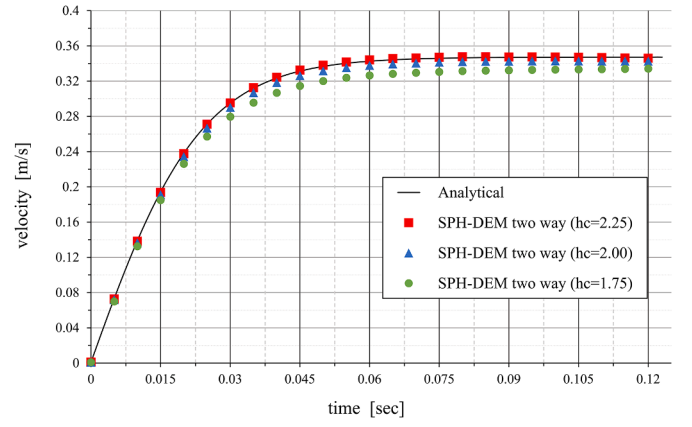


Fig. 13. Terminal velocity of single DEM particle.

solid particle velocity and the undisturbed fluid velocity evaluated at the solid particle location. This premise is valid for one-way coupling between SPH and DEM, since the behavior of DEM particles does not disturb the flow field of SPH (Gualtieri et al., 2015; Horwitz & Mani, 2016). However, in the case of two-way coupling, the fluid velocity is affected by the neighbor DEM particles due to the momentum exchange term S_f in Eq. (28). This causes the estimated drag force to be underestimated, so that the terminal velocity is overestimated compared to the analytically solved value in two-way coupling, as shown in Fig. 14. Therefore, for the accurate SPH-DEM simulation, it is necessary to estimate the drag force error caused by using the disturbed fluid velocity and correct the drag force as a function of the DEM particle size to the SPH spacing. As a result of performing parametric study on SPH-DEM coupling resolution, the difference in terminal velocity values between one-way coupling and two-way coupling is estimated to be less than 0.5% when $\Delta x_{SPH}/d > 3$, as shown in Fig. 14.

5.2. Pressure drop through a packed bed

Flows through a packed bed are frequently encountered in various engineering fields (Li et al., 2018; Tallmadge, 1970). The range of applications associated with the process industry includes packed bed fluidized reactors and scrubbing and filtration. Moreover, in the field of nuclear engineering, the convective flow through a packed debris bed pertains to a major safety issue as it directly affects the coolability of debris particles in the late phase of core disruptive accidents (Li et al., 2017). To enhance the safety and reliability of these industrial technologies, it is necessary to model the flow through the packed bed.

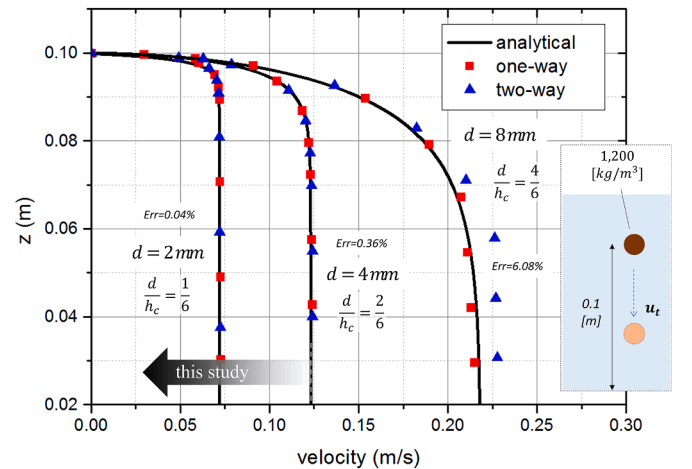


Fig. 14. Effect of coupling length for terminal velocity simulation.

Generally, the pressure drop or flow resistance is evaluated by applying empirical correlations, rather than performing direct numerical simulations due to the geometric nature of the packed bed and complexity of the flow through such beds. In this study, as indicated in Eq. (32), the flow resistance is evaluated by multiplying the drag force equations by the voidage term, which is a function of the local porosity. In this context, it is necessary to evaluate whether the current drag model could well reproduce the flow resistance of the packed bed system and whether the two-way coupling between SPH and DEM is valid.

The drag force applied in packed bed particles can be transformed and expressed as in the following pressure drop equation:

$$-\frac{dP}{dz} = \frac{3}{4d} C_d \rho |J| J (1 - \varepsilon) \varepsilon^{-1-\chi} \quad (40)$$

where ε is the averaged local porosity, and J denotes the relative superficial velocity. To evaluate the validity of the adopted drag force estimation method, the numerically estimated pressure drop (using Eq. (40)) is compared with that obtained using the Ergun equation (Ergun & Orning, 1949), which is the most widely used semi-empirical model for the pressure drop in porous media.

$$-\frac{dP}{dz} = \frac{150(1 - \varepsilon)^2 \mu}{d^2 \varepsilon^3} J + \frac{1.75(1 - \varepsilon) \rho}{d \varepsilon^3} |J| J \quad (41)$$

As shown in Fig. 15, the results obtained using Eqs. (40) and (41) exhibit the same tendencies for two different solid particle diameters ($d = 1.5$ [mm], $d = 6.0$ [mm]). The pressure drop increases as the flowrate increases, and the pressure drop in smaller particles ($d = 1.5$ [mm]) is relatively higher than that in larger bed systems ($d = 6.0$ [mm]). Furthermore, the results estimated through the SPH-DEM coupled simulation are in agreement with those obtained using Ergun's correlation.

A sensitivity study on the coupling domain (h_c) is performed in order to demonstrate the validity of two-way phase coupling between SPH and DEM for drag model. As shown in Fig. 16, it can be seen that the numerically estimated pressure gradient shows a good agreement with the analytic value calculated based on Ergun equation, especially for the large h_c . The result for quantitative evaluation of numerical errors is shown in Fig. 17. If h_c/D is greater than 4.0, the average error is estimated to be less than 0.5%, while the maximum error is less than 1.0%. In all SPH-DEM coupled simulations in this study, $h_c/D > 4.0$ was satisfied.

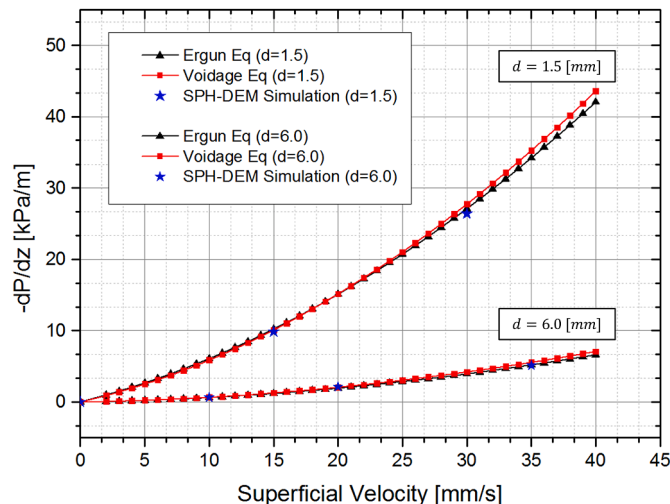


Fig. 15. Pressure drop of flow through packed bed (simulation results).

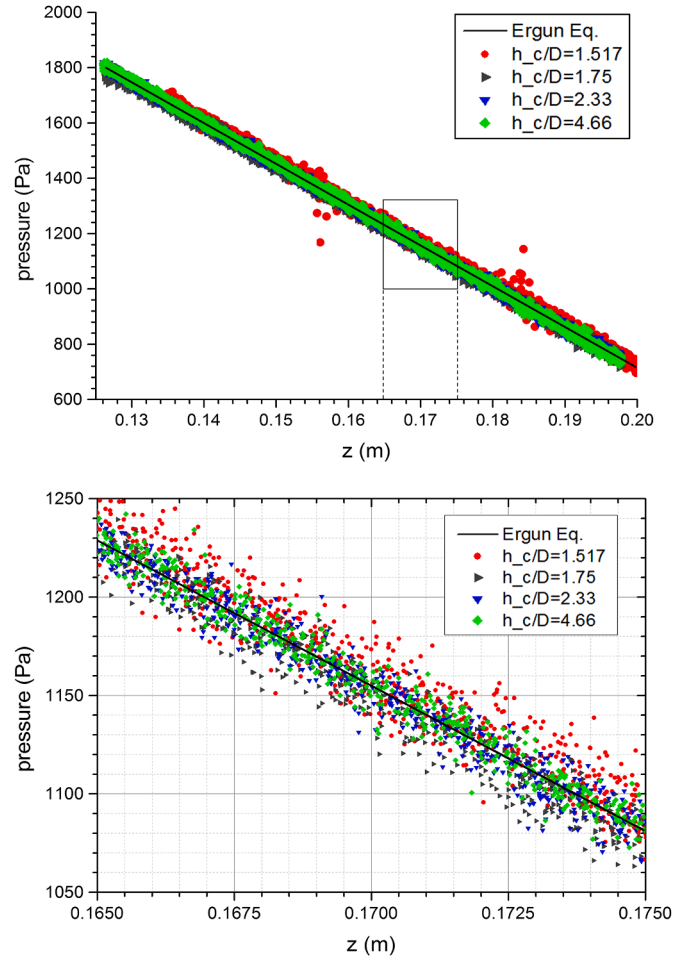


Fig. 16. Results of sensitivity study on coupling domain for pressure drop.

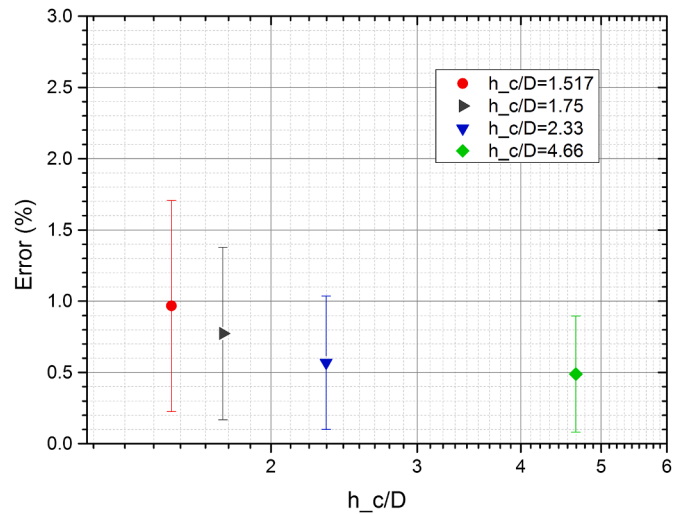


Fig. 17. Evaluation of numerical errors on pressure drop problem.

5.3. Particle-fluid dam-break

A 3D particle-fluid dam-break is simulated and compared with the benchmark experiment to validate the SPH-DEM coupled solver for the dynamic particulate flow involving free surfaces. The dam-break simulation is generally a reasonable test case to demonstrate the capability of a CFD-DEM coupled code as the flow pattern is dynamic and the local

porosity varies according to the velocity and distribution of the solid particles. In this kind of dynamic particulate flow, some solid particles may exhibit single particle behavior within one time-step, while others may be subjected to large porosity conditions. In this respect, it is necessary to evaluate whether the drag model well predicts the overall particulate flow even in dynamic problems with wide range of porosity.

The benchmark experiment reported by Sun (2013) (X. Sun et al., 2013) is implemented by filling the water on one side of a water tank and removing the gate with a specific velocity at a certain time. The water tank has dimensions of $200 \times 150 \times 150$ [mm] and is split by the gate located 50 mm from one side of tank wall. The water column with a depth of 100 mm collapses as the moving gate is removed (0.68 m/s). In the case of the solid particle phase, spherical particles with a total mass of 200 g are initially generated and settle behind the moving gate. A total of 182,716 SPH particles are generated for the water column, and 7,762 spherical DEM particles are produced for solid particles. The physical properties of each phase and simulation conditions are summarized in Fig. 18 and Table 1.

The simulation results of the SPH-DEM coupled solver are qualitatively and quantitatively validated considering the experimental results. Fig. 19 shows the qualitative validation results compared to the experimental snapshots at the same time interval. The behaviors of both the fluid and solid phases in the benchmark experiment are well reproduced in the SPH-DEM coupled simulation. The quantitative validation is performed by comparing the propagation of the leading front with the experimental data for both the fluid and solid particle phases. The following nondimensional parameters are defined for the quantitative validation:

$$x^* = x/a \quad (42)$$

$$t^* = t'(2g/a)^{0.5} \quad (43)$$

where x and t' denote the leading front and time, respectively, and the asterisk indicates the nondimensional form of each variable. a denotes the width of the initial water column, as shown in Fig. 18, and g represents the gravitational acceleration. Fig. 20 shows the quantitative validation result for the particle-fluid dam-break simulation. The SPH-DEM simulation results match with those of the benchmark experiment for both the fluid and solid particles.

Additionally, a decoupled simulation and one-way coupled simulation are performed for the dam-breaking behavior to evaluate the momentum exchange between the SPH and DEM. In the case of decoupled simulation, no force acts between the SPH and DEM particles. Thus, the fluid exhibits normal dam-breaking behavior for a single-phase, and the solid particles exhibit bead-like behavior. In the case of one-way coupled simulation, only the DEM particles move under the influence of the fluid, whereas the SPH particles do not exhibit any reaction forces for the solid particles. In this case, the fluid phase still exhibits a normal dam-breaking motion, whereas the solid particles follow the fluid under the

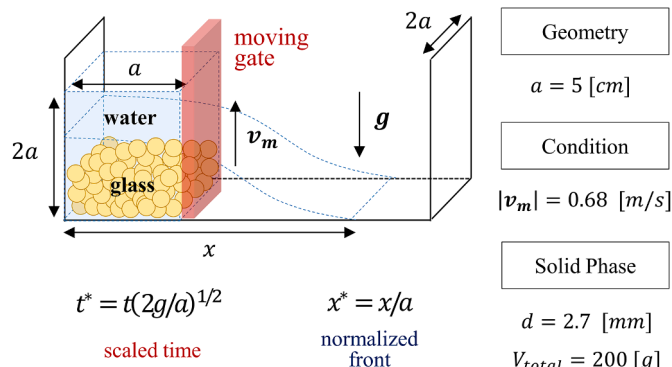


Fig. 18. Geometry and conditions for fluid-solid dam break simulation.

Table 1

Modelling parameters and conditions for fluid-particle dam-break simulation.

Fluid (Water)	Density [kg/m ³]	1,000
	Dynamic viscosity [Pa·s]	0.00089
Solid Particles	Density [kg/m ³]	2,500
	Young's modulus [Pa]	1.0e ⁸
	Restitution coefficient	0.9
	Friction coefficient	0.2
	Total mass [kg]	0.2
Simulation Conditions	# of SPH particles	182,716
	# of DEM particles	7,762
	Time step	2.0e ⁻⁶

action of the drag force. The results for the three different momentum exchange methods indicate that only the two-way phase coupled model simulates the actual physics, and the one-way coupled simulation overestimates both the fluid and solid particle phases, as shown in Fig. 21. The results of the decoupled and one-way coupled simulation for the fluid phase are rather in agreement with the experimental results of the single fluid dam-breaking motion, as shown in Fig. 20.

5.4. Gas-driven three-phase leveling behavior of spherical debris

To evaluate the capability of the SPH-DEM coupled solver for gas-liquid-solid three-phase flow, the numerical simulation for the vapor-driven leveling behavior of spherical solid particles is performed, which is a key phenomenon associated with the coolability of solid debris in the nuclear engineering field. In the late phase of a severe accident of pressurized water reactor, the core melt may be released from the reactor vessel if the ex-vessel cooling cannot sufficiently remove the decay heat from the core melt (Fig. 22). In this case, the core melt is jetted into a water pool and may fragment into solid particles depending on the depth of the wet cavity. These debris particles may spread via coolant drag and finally sediment into a mound shape, as shown in Fig. 22. In this configuration, coolant boiling and subsequent two-phase natural convection occur due to the decay heat of the debris particles. Under the influence of this convective flow, the conical shape of the particulate debris bed may change, and the bed may be leveled, as shown in Fig. 23. This self-leveling phenomenon is a critical safety issue in terms of the mitigation of the severe accident as such behavior may change the non-coolable configuration to a coolable geometry by reducing the possibility of a dry-out (Basso et al., 2016).

To examine this vapor-driven leveling behavior, several series of experiments have been elaborately designed and performed in Kyushu University to clarify the mechanisms underlying the self-leveling behavior of debris beds (CHENG et al., 2011; Cheng et al., 2013, 2014). The tendency of leveling behavior has been analyzed under various experimental conditions, and several empirical models have been developed to describe the transient variation in the bed angle during the self-leveling behavior. Three different gas generation methods (i.e., gas injection, depressurized boiling, and bottom-heated boiling) have been considered to simulate the vapor-driven leveling behavior. Among these methods, in this study, the gas injection experiment in a rectangular tank is considered as a benchmark case to validate the SPH-DEM solver. A rectangular viewing tank ($500 \times 250 \times 55$ [mm]) is filled with 2.5 L of water and glass bead particles ($20^\circ - 27^\circ$ inclined) with a diameter of 6 mm. The depth of the water column is approximately 400 mm, and nitrogen gas is released into the tank from the bottom at a specific gas flow rate (1.53, 3.43, 4.16 [L/min]). A porous medium known as airstone is placed at the tank bottom to serve as a gas distributor and ensure that uniform gas percolation is realized. The experimental conditions for the cases considered for the code validation are summarized in Fig. 24 and Table 2.

The numerical simulation of the SPH-DEM solver is performed under the same conditions as in the experiment cases. The physical properties of the fluid and solid particles pertained to the real properties of purified

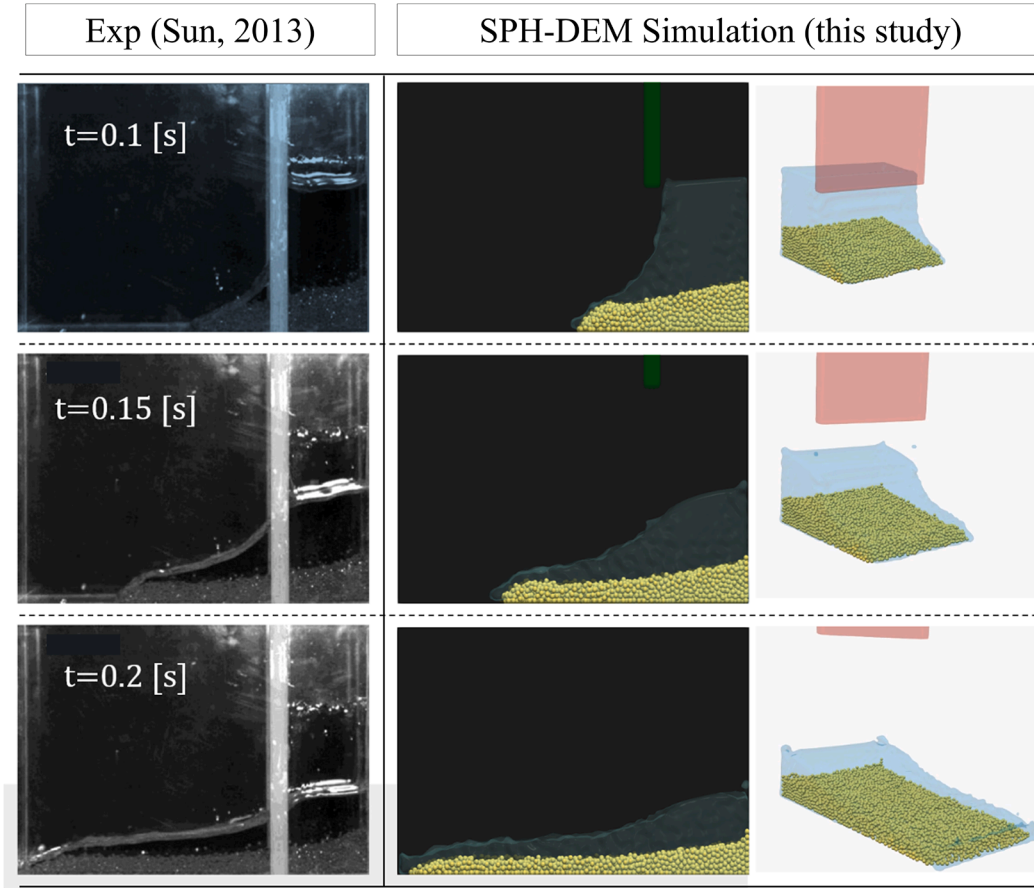


Fig. 19. Simulation results for fluid-solid dam-break behavior.

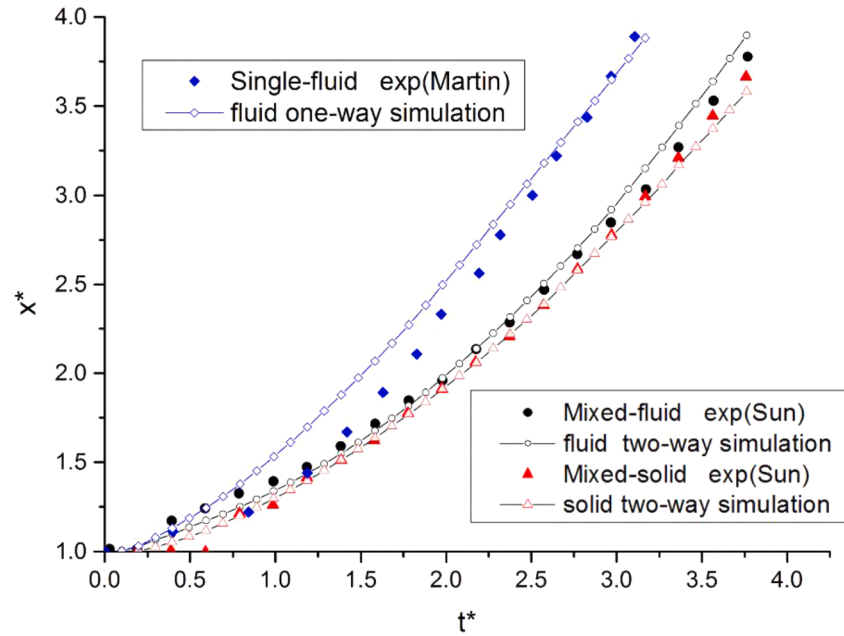


Fig. 20. Validation results of fluid-solid dam-break simulation.

water and glass beads, respectively. The incidental properties that were not described in the benchmark experiment are obtained by referring to the numerical analysis performed in the same institution (Guo et al., 2013). In the SPH-DEM coupled simulation, the spacing of the SPH

particles is set to be sufficiently fine to ensure a reasonably high resolution of the fluid domain, but not excessively fine to ensure the accuracy of momentum exchange between the SPH and DEM particles. The nitrogen SPH particles are generated in a checkerboard shape with a

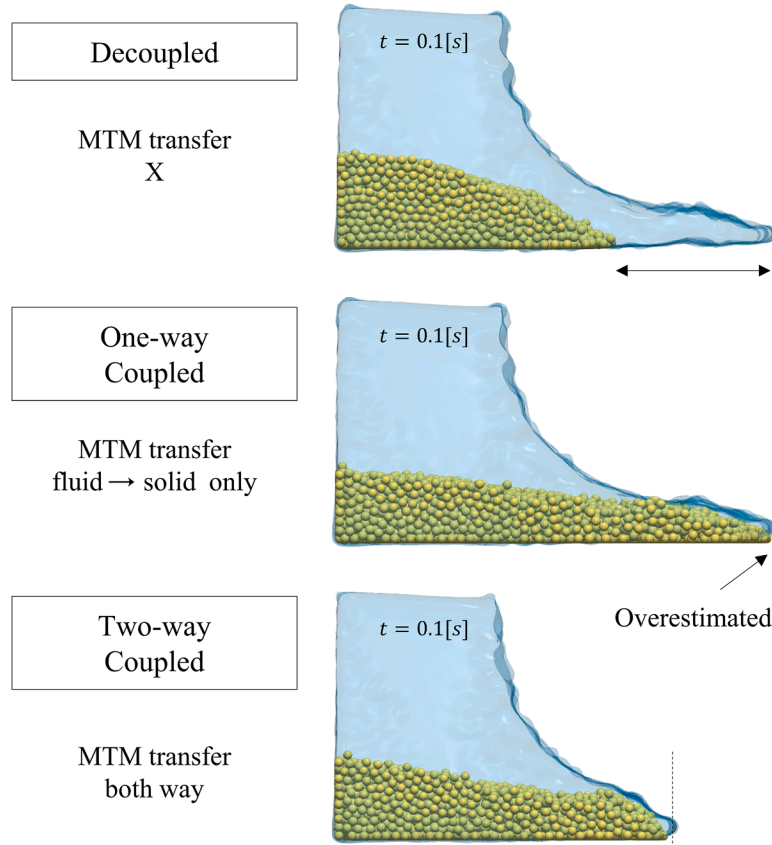


Fig. 21. Comparison results for momentum exchange way.

specific velocity to reproduce the gas injection method of the benchmark experiment using the porous medium (airstone), as shown in Fig. 25. The gas is injected after sufficient time for the liquid phase to reach the hydrostatic state and also for the solid particles to completely converge to a zero kinetic energy state. The physical properties and simulation conditions are summarized in Table 2.

To validate the SPH-DEM solver, the ratio of the inclination angle θ at a specific time t' to the initial angle θ_0 is estimated as

$$R(t') = \frac{\text{Inclination Angle at } t' \theta(t')}{\text{Initial Inclination Angle } \theta_0} \quad (44)$$

and compared with the experimental data. The tangent of the inclination angle at a specific time is determined as the ratio of the height of apex to the half-width of the tank both in the experiment and numerical simulation:

$$\theta(t') = \tan^{-1} \left(\frac{\text{Height of Apex, } H_{\text{apex}}(t')}{\text{Half Width of Tank, } W_{\text{tank}}} \right) \quad (45)$$

where H_{apex} and W_{tank} denote the maximum height of the apex and half-width of the rectangular tank, respectively. The height of the apex is determined for the DEM particles within a specific area, as shown in Fig. 25.

The SPH-DEM simulation results for a gas flow rate of 4.16 L/min are qualitatively compared with the snapshots of the experimental result, as shown in Fig. 26. When the gas starts to be injected, the rising bubble creates a wake of the following liquid, and the momentum of this wake physically moves the surface solid particles. This series of leveling behavior were clearly observed in SPH-DEM coupled simulations, which is also identified through various experiments (Jasmin Sudha et al., 2015; Cheng et al., 2013; Teng et al., 2021). The leveling behavior of the particle bed can be observed in both the experiment and simulation

results. Fig. 27 shows the quantitative comparison of the $R(t')$ value obtained in the experiment and simulation. The SPH-DEM results well predict the overall tendency of the leveling behavior of the debris bed.

Although the solid particles are relatively static in this leveling problem, it can be seen that the capillary effect due to the channel constriction are pretty well reflected through the surface tension model and following momentum exchange, result in good agreement in leveling tendency with the benchmark experiment.

6. Summary and conclusion

A gas-liquid-solid three-phase flow analysis solver was constructed based on a two-way phase coupling of a DEM model and multiphase SPH method to be applied in various fields. The particle-based SPH method was adopted for the fluid phase to exploit its capability for solving multiphase flows involving a sharp and dynamic interface between the gas and liquid, and the DEM was adopted to track the behavior of solid particles. The coupling between the SPH and DEM methods was implemented in an unresolved manner for numerous applications associated with the particulate flow and packed bed systems. To enable the use of the proposed SPH-DEM system in practical applications, the SPH-DEM solver was parallelized using a GPU, and the governing equations and momentum exchange equations were solved concurrently for each SPH/DEM particle with the appropriate parallelization algorithms. The GPU parallelization was quantitatively evaluated through a simple scalability analysis, and the results indicated that the performance of the accelerated version was considerably enhanced, especially for large-scale simulations.

To demonstrate the capability and applicability of the proposed approach in various multiphase applications, several benchmark simulations were performed. The terminal velocity and packed bed pressure drop simulation demonstrated the validity of the momentum exchange

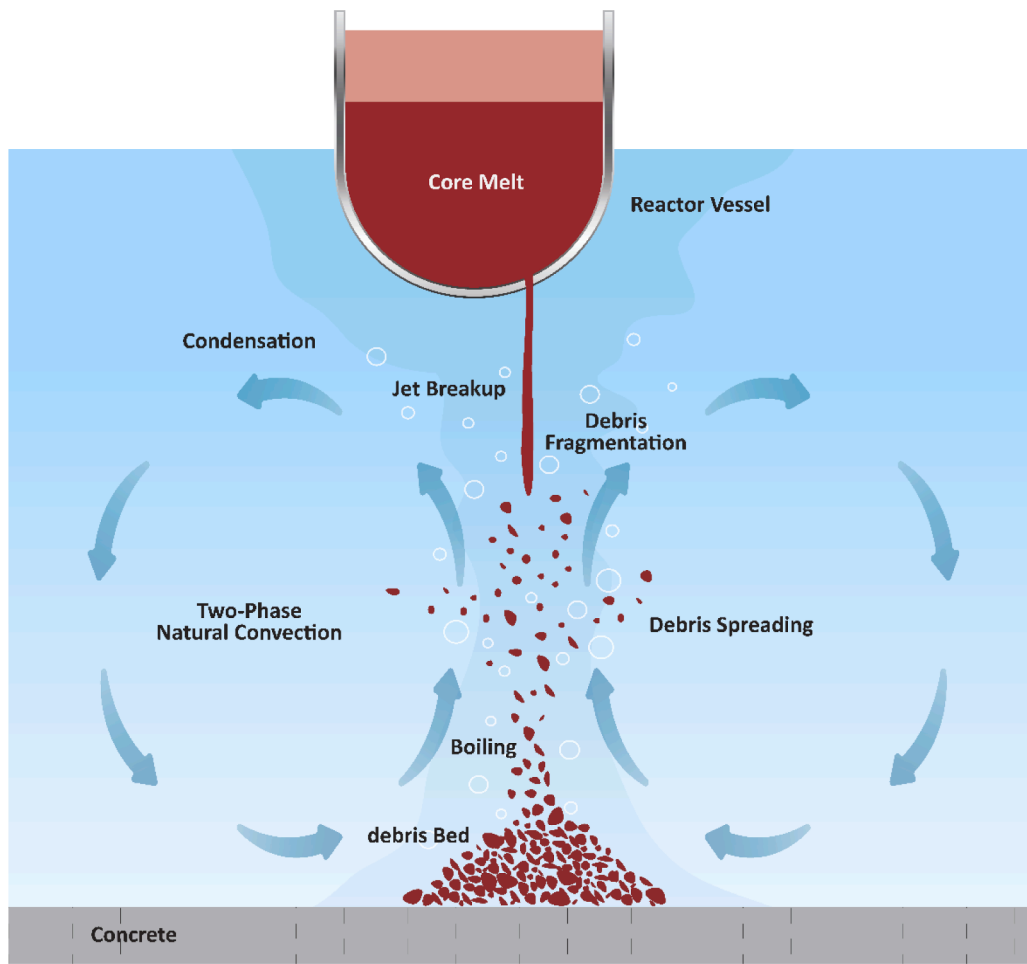


Fig. 22. Phenomena on the late phase of severe accident of nuclear reactor.

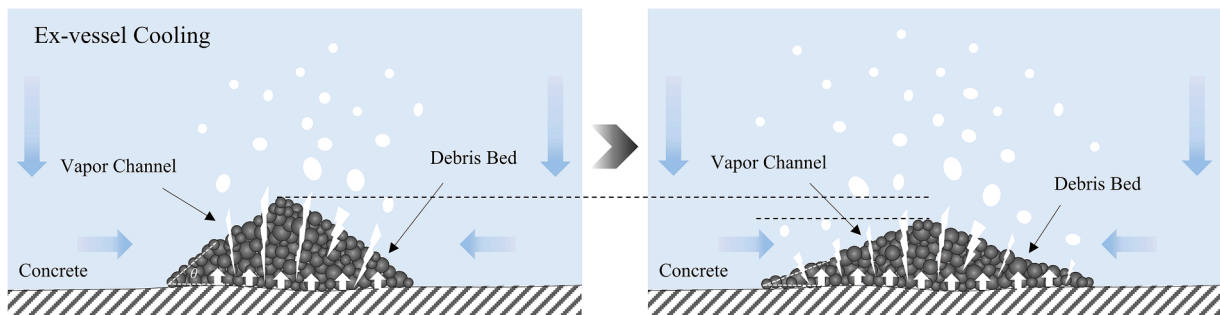


Fig. 23. Self-leveling behavior of debris bed.

model between the SPH and DEM. The feasibility of the SPH-DEM solver for particulate flow was demonstrated through a particle-fluid dam-break simulation involving free surfaces. In this simulation case, the importance of two-way phase coupling in terms of momentum exchange was also highlighted. Furthermore, the gas-driven leveling behavior of spherical particles was simulated as a case study to demonstrate the applicability of the constructed SPH-DEM solver for gas-liquid-solid three-phase flow. The results of the simulation were in good agreement with those of the corresponding benchmark experiment.

In the considered numerical simulations, the potential and applicability of the SPH-DEM three-phase flow solver was demonstrated, albeit within a limited range (i.e., hydrodynamic behavior for incompressible three-phase flows). In addition, although the capability of the model for three-phase flow analysis has been partially demonstrated through this

vapor-driven leveling problem (section 5.4), the simulation cases performed in this study mostly focused on the static conditions for particle bed due to various limitations on validation. Since the SPH-DEM coupled model proposed in this study is expected to an effective approach to dealing with particulate behavior within sharp and dynamic two-phase flows, additional V&V for dynamic three-phase flow is required in which the surface tension behavior of solid particles can be clearly observed. Unfortunately, however, there are few benchmark data which can be used for V&V of proposed SPH-DEM coupled model. Specifically, although there are several experiments that observed the three-phase particulate flow behavior (multi-phase sedimentation (Konovalenko et al., 2016), vapor-driven sloshing (Cheng et al., 2019), etc.), the computational cost is too expensive to reproduce those experiments with the single-GPU parallelized solver. Thus, it seems that

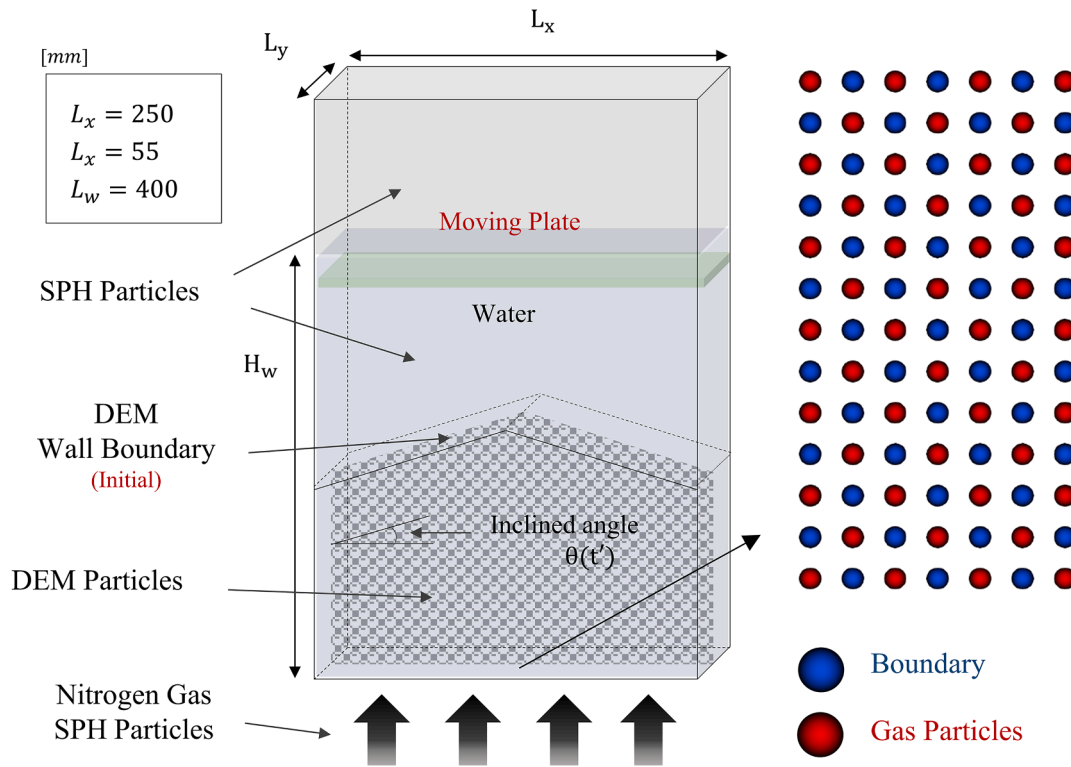


Fig. 24. Geometry and conditions for leveling simulation.

Table 2

Modelling parameters and conditions for vapor-driven leveling simulation.

Liquid phase	fluid	Purified water
	Density [kg/m^3]	1,000
	Dynamic viscosity [$\text{Pa}\cdot\text{s}$]	0.001
Gas Phase	fluid	Nitrogen gas
	Density [kg/m^3]	1,126
	Dynamic viscosity [$\text{Pa}\cdot\text{s}$]	1.789e^{-5}
Solid Particle Phase	Flow rate [L/min]	1.53, 3.43, 4.16
	solid	Glass bead
	Density [kg/m^3]	2,590
	Poisson's ratio	0.25
	Elasticity modulus [Pa]	1.0e^{10}
	Friction coefficient	0.3
	Restitution coefficient	0.9
	Particle diameter [mm]	6.0
	Initial inclined angle [$^\circ$]	20~27
	# of SPH particles	231,664
Simulation Conditions	# of DEM particles	12,480
	Time step	2.0e^{-6}

there are many limitations in practical quantitative validation of proposed SPH-DEM coupled model. In these reasons, further works are required on the following items:

- Various high-performance computing (HPC) techniques should be employed. For the three-dimensional validation and application of the proposed SPH-DEM coupled model, additional parallelization and acceleration should be made by applying various HPC algorithms.
- Additional V&V and further experimental studies for V&V are required for various three-phase flow conditions from single phenomenon to dynamic and complicated problems. The validation was performed mostly focused on the static conditions for particle bed due to several limitations, which is partial validation of proposed SPH-DEM model. Thus, additional works for further V&V are required

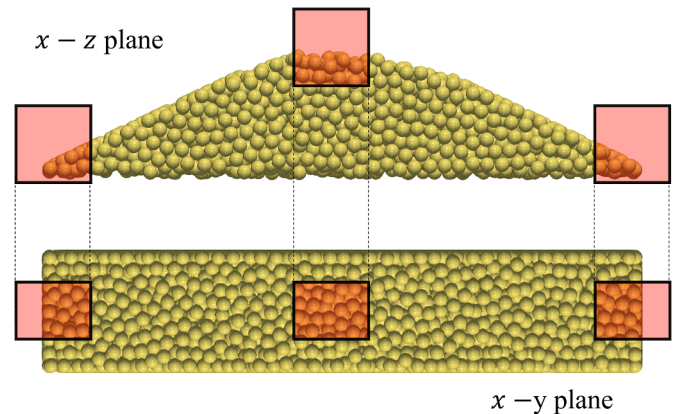


Fig. 25. DEM particles used for data analysis.

- Physical models for heat transfer, phase change, etc. should be additionally implemented so that the proposed numerical model can have the capability for deeper coupling between multiphase flow and solid particles. Appropriate V&V of each physical model is also required.

CRediT authorship contributino statement

Young Beom Jo developed and accelerated the SPH-DEM coupled code and performed simulations, analyzed data, and drafted/revised the manuscript. So-Hyun Park co-developed the SPH model of the code. Hee Sang Yoo improved the SPH-DEM coupled model and participated in revision. Eung Soo Kim supervised the research, and provided the revision of paper.

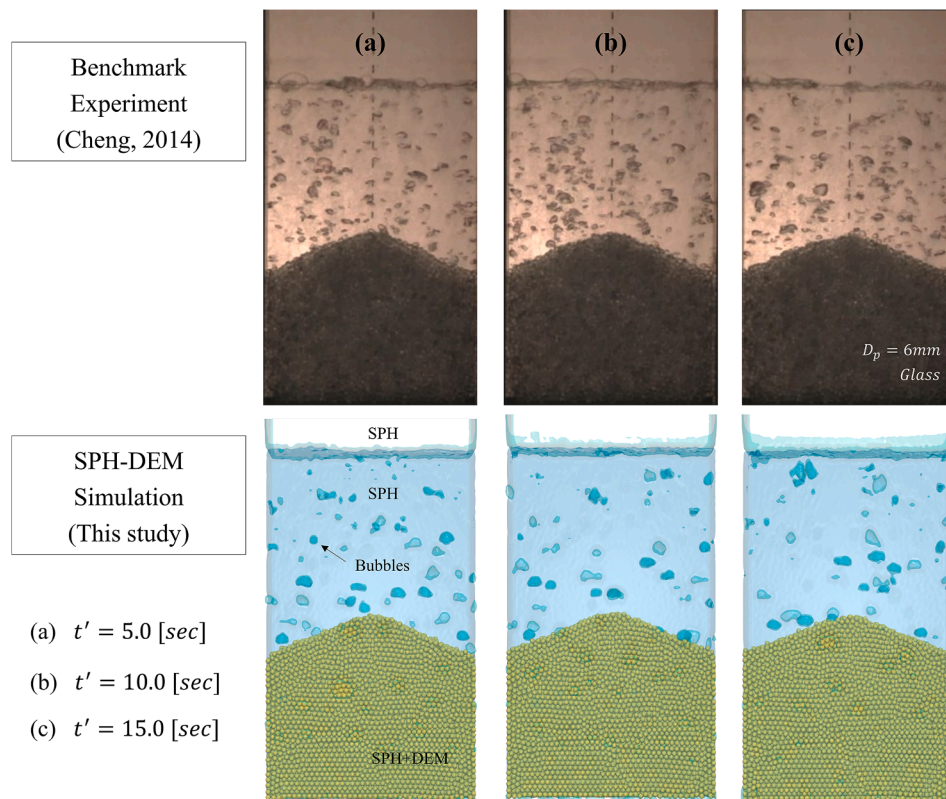


Fig. 26. Comparison of simulation results with experimental snapshots.

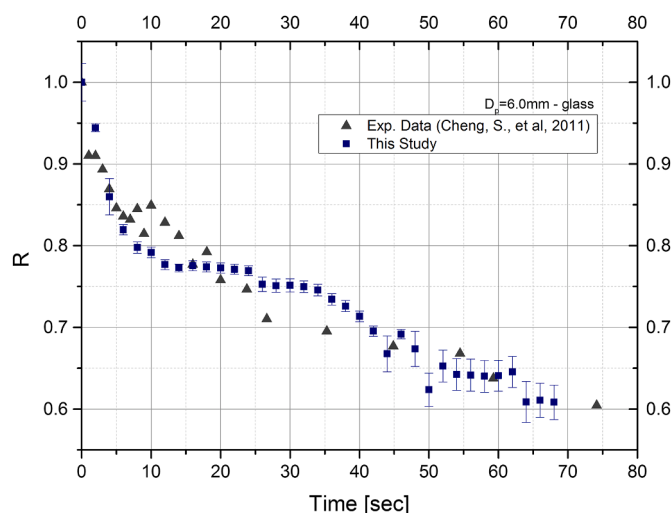


Fig. 27. Validation results for R value.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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